Coefficient Extraction Operators in the Theory of Multidimensional Random Walk Enumeration

by

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B.Sc. Mathematics, Simon Fraser University, 2012

A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF

Master of Science

in

THE FACULTY OF SCIENCE
(Mathematics)

The University of British Columbia
(Vancouver)

April 2015

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Abstract

In this work we explore the the application of operators to enumeration of multidimensional random walks with a focus on walks which interact with the boundary hyperplanes. Using operators makes the derivation of the functional equations a lot easier and some preliminary work shows that using operators is the way to go when solving problems of this nature.

Up to 350 words

Preface

This work intends to establish a new framework of notation and tools for the enumeration of random walks with or without interactions, and similar problems. The notation makes it much easier to explore the symmetries of the problem and write down summation formulas where it was not possible to do so before (with a simplified notation). The tools make the derivation of functional equations for various problems an almost trivial matter. The real power is in combining both approaches to gain insights into the generating function world of enumerating random walks on constrained regions in multidimensional spaces.

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Glossary

This glossary uses the handy `acroynym` package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the LaTeX source.

Acknowledgments

I wish to say thank you to everyone who has helped me along my graduate school journey.

Thank you Victoria Zhu for supporting me throughout all these years. Without you love and support I would have quit many times! You were always there for me when I had doubts and wanted to give up.

Thank you Andrew Rechnitzer for taking me under your wing and believing in me. You are the man!

Thank you Kalle Karu. Without your support I wouldn't have made it this far. Your generousity will be forever appreciated!

Thank you mom and dad for believing in me, guiding me and supporting me in all the times of need. There's nothing quite like hearing you say that I can do it!

Thank you Tony Chong, Gloria Chien and Vlad Bohdanov for being my coaches and mentors, and helping me on the path to become a champion!

Also a big thank you to all my friends for letting me rant so many times!

Last but not least I want to thank UBC and the mathematics department especially for supporting me through thick and thin and somehow making things work for me whenever all hope was lost.

If I forgot anybody...you can just kill me.

Chapter 1

Introduction

We'll investigate the generating functions for quarter-plane walks where we track how many times these walks touch the walls.

- Q is the generating function.
- Powers of variables x_i keep track of the i-th coordinate of the endpoint of a random walk. For example the monomial $x_1^3 x_2^1 x_4^7$ counts how many walks end at $(3,1,0,7)$.
- Powers of variables a_i keep track of the number of times a random walk passes through a point with i-th coordinate equal to zero. For example the monomial $a_1^2 a_2$ counts how many times the random walks have passed through points with coordinates $(0,j)$ and $(i,0)$ respectively.
- Powers of t count the length of random walks.
- Variables with overline are reciprocals like $\overline{x_1} = 1/x_1$.
- S is the Laurent polynomial generating function for the allowed step set. For example for the steps $\{\text{N, E, SW}\}$ we have $S = x_2 + x_1 + \overline{x_1 x_2}$.
- We'll often use $b_i = a_1 - 1$ and $c_i = b_i/a_i = 1 - \overline{a_i}$.

Next we have some notation on extraction of coefficients

- $[d] = (1,\ldots,d)$ as a sequence.

- For a sequence $I = (i_1, \ldots, i_n)$ and a sequence $P = (p_1, \ldots, p_n)$, $x_I^P = x_{i_1}^{p_1} \ldots x_{i_n}^{p_n}$. For example $x_{[3]}^{(1,0,2)} = x_1^1 x_2^0 x_3^2 = x_1 x_3^2$. We'll usually use this notation for when we have general sequences, not in the case of the example which we would write out explicitly.

- $[u_I^P]$ stands for extracting the coefficient infront of u_I^P in the generating function expansion.

- $Q_I^J = [x_I^P]Q$ and similarly $S_I^J = [x_I^J]S$. For example if $I = (1,3)$, $J = 0,1$, (omiting parentheses) $Q_I^J = Q_{1,3}^{0,1}$. We will sometimes write $(Q_I^P)_J^R = Q_{I,J}^{P,R}$

- For a set I, $\mathbb{O}$ and $\mathbb{I}$ stand for sequences of $|I|$ zeros and ones respectively. For example for $I \subseteq [d]$ we might have $S_I^{-\mathbb{I}} Q_I^{\mathbb{O}}$. Analogously we will write $(Q_I^{\mathbb{I}})_J^{\mathbb{O}} = Q_{I,J}^{\mathbb{I},\mathbb{O}}$

So we have

$$Q(x_1, x_2, a_1, a_2; t) = \sum_{i,j,k} Q_{1,2}^{i,j} x_1^i x_2^j t^k$$

So far we won't extract a's because we have an unobvious choice between extracting a, b or c with various pros/cons. For example it may sometimes be easier to extract b_1^k than a_1^k but there's the added cost that the extracted term counts walks with varying powers of k. We usually won't extract with respect to t.

Chapter 2

Introduction to Methods - 1D random walks

2.1 Derivation of the functional equation

In the interest of working out higher dimensional labelled walks' problems it is very useful to first visit the 1-dimensional case which will illustrate a lot of the methods that will later come into play. The 1-dimensional case will also give us a clearer picture as to how things work in higher dimensions (although the picture becomes more complicated). We'll treat the problem using generating functions. We'll use lower case letters and not quite the full notation because it is not necessary. In one dimension, on one axis, we'll use steps of length at most 1. If we have a random walk ending at coordinate i we can either step back adding -1 to the coordinate, step forward adding $+1$ to the coordinate or staying in one place, adding 0 to the coordinate. This can be somewhat formalized by using polynomials. Ending a walk at coordinate i will be encoded with a monomial x^i. Then stepping backwards, forwards or staying in one place will be denoted with x^{i-1}, x^i and x^{i+1} respectively. This is the same as multiplying by $1/x = \bar{x}, 1, x$ in each case. We can encode this information is a Laurent step polynomial s defined as

$$s = s_{-1}\bar{x} + s_0 + s_1 x$$

Here s_i will be either 0 or 1 for any specific i, tracking whether we have allowed those kinds of steps or not. Keeping this in mind, there are 8 possible step sets because we have two options for each s_i. We keep track of the length of a random walk with powers of t and we keep track of its endpoint with powers of x. We'll also label every interaction with the origin by multiplying by a every time a walk steps on the origin. We'll keep the walks on the non-negative axis.

Long story short we can find the generating function in a very simple, brute force manner, even for higher dimensions. We start with the empty walk denoted by 1 since it starts and ends on the origin without taking a step, so no t, x or label a. Then try out each step from s and eliminate those that take us past the origin in the negative direction. If we step on the origin we multiply by a. Those walks that remain are then multiplied by t since they are not of length 1. We can proceed iteratively k tiems to obtain random walks on the non-negative axis of length k. Adding up the results allows us to write the generating function up to $O(t^k)$. It will prove very useful to think about operators so we can formalize this notion in terms of operators. For example, here we are taking walks of length $k-1$, trying all options of s (this is like multiplying by s), removing all the walks that end up on the negative axis (this is like removing the $\bar{x}$ term) and labeling those that end on the origin (this is like multiplying the x^0 coefficient by a). Let D stand for the operator that cleans up the walks and labels the valid ones. Then if q_k stands for the labelled walks of length k, then the k-th term of q is $q_k t^k$ and we have

$$q_{k+1}t^{k+1} = tD(sq_k t^k)$$

Thus we can write the generating function q iteratively

$$q = \sum_{k=0}^{\infty} t^k (Ds)^k (1)$$

where we're iteratively applying Ds to 1 to obtain all labelled walks. We can treat this operator Ds as a formal variable and write

$$q = \sum_{k=0}^{\infty} (tDs)^k (1) = \frac{1}{1-tDs}(1)$$

and then this gives us a functional equation for q

$$(1 - tDs)q = 1$$

There is another way of deriving this equation while not treating Ds as a formal variable. Instead of applying the operator iteratively we can apply it on the generating function as a whole. The product tsq is the generating for all random walks that end with x-coordinate greater or equal to -1 where some steps end up on x-coordinate 0 without being labelled. We can apply D on this product to remove the terms ending at x-coordinate -1 and label those that end up on x-coordinate 0. This leaves us with all the valid walks encoded in the generating function q without the empty walk 1. Therefore we can write a functional equation for q as

$$q = 1 + D(tsq) = 1 + tDs(q)$$

and we ca re-write this as $(1 - tDs)q = 1$ as before. Now, we have to say a few words about what the operator D actually is. First we are removing all the walks that end on x-coordinate equal to -1 which can be handled by the operator $E = 1 - \overline{x}[\overline{x}]$ which removes those walks. Then the labelling can be handled by the operator $L = 1 + (a-1)[x^0] = 1 + b[x^0]$ where $b = a - 1$. Therefore we can write $D = LE$

$$\begin{aligned} D = LE &= (1 - \overline{x}[\overline{x}])(1 + b[x^0]) \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] \end{aligned}$$

since the last term vanishes on all generating functions. We could have also applied the operators in reverse to get the same result. Thiscommutativity makes sense because on the level of the function they are operator interacting with different

monomials. We can now expand the functional equation to the following

$$\begin{aligned}1 &= q - tLE(sq) = q - t(sq + b(sq)_0 - \overline{x}(sq)_{-1}) \\ &= q - t(sq + b(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - \overline{x}(s_{-1}q_0)) \\ &= q - t(sq + b(s_{-1}q_1 + s_0q_0) - \overline{x}(s_{-1}q_0))\end{aligned}$$

since $q_{-1} = 0$. We can rewrite this equation in the following form

$$(1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - bts_{-1}q_1 = 1$$

Here $q_0 = q(0)$ and $q_1 = q'(0)$ where $q = q(x)$. In the interest of simplifying this equation we can eliminate the q_1 term by extracting the x^0 coefficient from the whole equation

$$\begin{aligned}1 &= q_0 - t(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - tbs_0q_0 - bts_{-1}q_1 \\ &= q_0 - t(s_{-1}q_1 + s_0q_0) - tbs_0q_0 - bts_{-1}q_1 \\ &= (1 - ts_0 - tbs_0)q_0 - (ts_{-1} + bts_{-1})q_1 \\ &= (1 - ats_0)q_0 - t(s_{-1} + bs_{-1})q_1 \\ &= (1 - ats_0)q_0 - ats_{-1}q_1\end{aligned}$$

We can solve this equation for q_1

$$ts_{-1}q_1 = \overline{a}((1 - ats_0)q_0 - 1)$$

and substitute the result into the functional equation (for now we don't worry about the possibility of $s_{-1} = 0$ because it leads to the same thing anyway).

$$\begin{aligned}1 &= (1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - c((1 - ats_0)q_0 - 1) \\ &= (1 - ts)q - (c(1 - ats_0) + bts_0 - t\overline{x}s_{-1})q_0 + c \\ &= (1 - ts)q - (c - bts_0 + bts_0 - t\overline{x}s_{-1})q_0 + c \\ &= (1 - ts)q - (c - t\overline{x}s_{-1})q_0 + c\end{aligned}$$

and therefore we can rewrite the equation as

$$(1-ts)q-(c-t\bar{x}s_{-1})q_0=\bar{a}$$

since $1-c=1-(1-\bar{a})=\bar{a}$. This process is rather complicated, even though it can be done for higher dimensional problems as well (as you can see in various previous approaches). Also there is an issue that we need to consider every time we are solving an equation in order to substitute quantities back into the equation. This is where the operator notation comes to the resque. It turns out the the L operator is invertible with inverse given by

$$L^{-1}=M=1-c[x^0]$$

and we can apply it on the functional equation. The RHS is simply $M(1)=1-c=\bar{a}$ and the LHS is

$$\begin{aligned}M(1-tLEs)q&=Mq-tEsq\\&=(q-cq_0)-t(sq-\bar{x}s_{-1}q_0)\\&=(1-ts)q-(c-t\bar{x}s_{-1})q_0\end{aligned}$$

and we've derived the same equation but in a much neater way. This will prove crucial later on when we derive a functional equation for higher dimensional problems. Although it can be done directly with a lot of effort it's much better to just use the operator notation. The derivation is brought down from about a dozen pages to one or two. The operator notation will also allow us to solve some multidimensional problems directly or with the help of lower dimensional problems.

We can extract the x^0 coefficient from the final functional equation again and although we can do this directly, we'll use the operator notation to tidy up the process. This will show up later when dealing with higher dimensional problems

$$\begin{aligned}[x^0]M&=[x^0]-[x^0]c[x^0]=[x^0]-c[x^0]=\bar{a}[x^0]\\[x^0]E&=[x^0]-[x^0]\bar{x}[\bar{x}]=[x^0]\end{aligned}$$

Then we extract the x^0 from the functional equation

$$[x^0]\overline{a} = \overline{a} = [x^0](M - tEs)q = ([x^0]M - t[x^0]Es)q = (\overline{a}[x^0] - t[x^0]s)q = [x^0](\overline{a} - ts)q$$

Therefore the extraction leads to

$$[x^0](1 - ast)q = 1$$

We could extract higher coefficients from the functional equation too. Let $k \geq 1$. Then

$$[x^k]M = [x^k] - [x^k]c[x^0] = [x^k]$$
$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]$$

Therefore

$$[x^k]\overline{a} = 0 = [x^k](M - tEs)q = ([x^k]M - t[x^k]Es)q = ([x^k] - t[x^k]s)q = [x^k](1 - ts)q$$

That is

$$[x^k](1 - ts)q = 0$$

and we can readily extract whichever coefficients we want. The first equation leads to

$$(1 - ats_0)q_0 - ats_{-1}q_1 = 1$$

and the second leads to

$$q_k - t(s_{-1}q_{k+1} + s_0q_k + s_1q_{k-1}) = 0$$
$$(1 - s_0t)q_k - t(s_{-1}q_{k+1} + s_1q_{k-1} = 0$$

2.2 Solving the functional equation for various s

In this section we'll solve all 8 one-dimensional problems fully. Some cases are easier to deal with than others. As we'll see later in the higher dimensional cases, some problems are easierto solve than others. This difficulty tends to scale up with

the inclusion of certain steps with negative direction.

2.2.1 The $s_{-1} = 0$ case.

This condition allows us to solve explicitly for q_0.

$$q_0 = \frac{1}{1 - as_0t}$$

We can substitute this expression in the functional equation and obtain

$$q = \frac{\overline{a}}{1 - st}\left(1 + \frac{b}{1 - as_0t}\right) = \frac{1 - s_0t}{(1 - st)(1 - as_0t)}$$

This takes care of half of the cases! What happens when we set $a = 1$, corresponding to setting $b = 0$ or extracting the b^0 coefficient from the equation? This corresponds to unlabelling the random walks. We obtain

$$[b^0]q = \widehat{q} = \frac{1 - s_0t}{(1 - st)(1 - s_0t)}$$

We'll note that

$$(1 - as_0t)q - (1 - s_0t)\widehat{q} = 0$$

This can also be written in operator form with $B = 1 - [b^0]$

$$B((1 - as_0t)q) = 0$$

This relates the labelled generating function to the unlabelled generating function! it says we can obtain the labelled generating function from the unlabelled one by the following relation

$$q = \frac{1 - as_0t}{1 - s_0t}\widehat{q}$$

We can also establish such as relationship for q_0 although the derivation here will be trivial. Extracting $[x^0]$ from the equation leads to

$$B((1 - as_0t)q_0) = 0$$

and

$$q_0 = \frac{1 - as_0 t}{1 - s_0 t} \widehat{q_0}$$

The relationship between q and $\widehat{q}$ is no accident but it tends to happen fairly often in the higher dimensional cases that we've considered. The functional equation in the $s_{-1} = 0$ case is easier to solve most likely because when $s_{-1} = 0$, walks don't go beyond the origin in the negative direction. Consequently the E operator acts as the identity on the generating function. and the functional equation reads as

$$(M - ts)q = q - cq_0 - tsq = (1 - ts)q - cq_0 = \overline{a}$$

2.2.2 The $s_{-1} \neq 0$ case.

The functional equation in the $s_{-1} \neq 0$ is also not difficult to solve. We'll try to apply the kernel method whenever possible. It requires finding an x value that makes the kernel $1 - ts$ equal to zero in the functional equation

$$(1 - ts)q - (c - t\overline{x})q_0 = \overline{a}$$

Such a solution will exist since $s_{-1} = 1$ and therefore the equation $1 - ts = 0$ is non-trivial. Let one such solution be $x = \sigma$ and suppose that we can plug it into the functional equation. Then the kernel vanishes, taking q away with it and we obtain

$$-(c - t\overline{\sigma})q_0 = \overline{a}$$

We can solve for q_0 explicitly again

$$q_0 = \frac{\overline{a}}{\overline{\sigma}t - c}$$

We can now solve for q

$$q = \frac{\overline{a}}{1 - ts}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - ts}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)$$

We note that we can apply the B operator again

$$B((a\overline{\sigma}t-b)q)=0$$

or written out explicitly

$$(a\overline{\sigma}t-b)q-(\overline{\sigma}t)\widehat{q}=0$$

This also allows us to write the labelled generating function in terms of the unlabelled generating function

$$q=\frac{\overline{\sigma}t}{a\overline{\sigma}t-b}\widehat{q}=\frac{t}{at-b\sigma}\widehat{q}$$

We can also work with q_0 at this stage. We find

$$[x_0]B((a\overline{\sigma}t-b)q)=B((a\overline{\sigma}t-b)q_0)=0$$

too. This ends up being the same as

$$(a\overline{\sigma}t-b)q_0-(\overline{\sigma}t)\widehat{q_0}=0$$

with solution

$$q_0=\frac{t}{at-b\sigma}\widehat{q_0}$$

The final task for this section will be to find what σ will be for various cases. We already know $s_{-1}=0$ so the result depend on s_0 and s_1. In general the kernel is zero when

$$\begin{aligned} ts &= 1 \\ t(s_{-1}+s_0x+s_1x^2) &= x \\ ts_1x^2-(1-s_0t)x+ts_{-1} &= 0 \end{aligned}$$

When $s_1\neq 0$ this is quadratic with two roots that are functions of t. We have to pick the one which has no singularity at $t=0$. When $s_1=0$ we have a linear equation which may have a single root. When $s_1=0$ we have the equation $(1-s_0t)x=t$ so

the single root is

$$\sigma = \frac{t}{1 - s_0 t}$$

When $s_1 = 1$ we have the equation $tx^2 - (1 - s_0 t)x + t = 0$ with roots

$$\sigma = \frac{1 - s_0 t - \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}, \quad \sigma^* = \frac{1 - s_0 t + \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}$$

The two roots multiply to 1 so $\overline{\sigma} = \sigma^*$. Later we shall have a little table tabulating all the results. Note that in the relationship between the labelled and unlabelled q_0 we have

$$q_0 = \frac{t}{at - b\sigma} \widehat{q_0}$$

where σ is a function containing a square root, i.e. not a polynomial. However we can go at the problem from a different angle to obtain a polynomial $f(a,t)$ for which $B(fq_0) = c$ which is not quite zero but it's a simpler relationship. This will also illustrate the group approach together with the kernel method. The two cases that we need to work on both have $s_{-1} = s_1 = 1$ with s_0 being 0 or 1. This means that s is left invariant under sending $* : x \mapsto \bar{x}$. We can apply this on the functional equation and now have a system of two functional equations

$$(1 - ts)q - (c - t\bar{x})q_0 - \bar{a} = 0$$
$$(1 - ts)q^* - (c - tx)q_0 - \bar{a} = 0$$

We can multiply the first by $c - tx$, the second by $c - t\bar{x}$ and subtract the equations to eliminate the q_0 term. We can divide everything out by $1 - ts$ and finally obtain

$$(c - tx)q(x) - (c - t\bar{x})q(\bar{x}) - \frac{(c - tx - c + t\bar{x})}{1 - ts}\bar{a} = 0$$
$$(c - tx)q(x) - (c - t\bar{x})q(\bar{x}) - \bar{a}\delta = 0$$

where

$$\delta = \frac{t(\bar{x} - x)}{1 - ts}$$

and crucially it doesn't depent on a. We extract the x^1 coefficient from the equation to obtain an equation relating q_1 and q_0

$$cq_1 - tq_0 - \overline{a}\delta_1 = 0$$

We can set $a = 1$ here so we get

$$-\delta_1 = t\widehat{q_0}$$

and we can bring this back to the original equation to get

$$\begin{aligned} cq_1 &= tq_0 - \overline{a}t\widehat{q_0} \\ bq_1 &= t(aq_0 - \widehat{q_0}) \end{aligned}$$

Earlier we had the equation

$$(1 - as_0t)q_0 = 1 + as_{-1}tq_1 = 1 + atq_1$$

which we can solve for q_1 this time and insert the solution into the above equation

$$b((1 - as_0t)q_0 - 1) = t^2a(aq_0 - \widehat{q_0})$$

Collecting coefficients and simplifying leads us to

$$\begin{aligned} (b(1 - as_0t) - a^2t^2)q_0 - (-at^2)\widehat{q_0} &= b \\ (c(1 - as_0t) - at^2)q_0 - (-t^2)\widehat{q_0} &= c \end{aligned}$$

which gives $f(a,t) = c(1 - as_0t) - at^2$ so that

$$B((c(1 - as_0t) - at^2)q_0) = c.$$

With two different approaches we obtained slightly different relationships upon applying the operator B.

2.2.3 1-dimensional case conclusion

In this section we were able to find explicit solutions for q_0, q and also B-operator relationships for q and $\widehat{q}$. In higher dimensions this may not be quite possible but we should still be able to find B-operator relationships for the equivalent of q_0 and $\widehat{q_0}$. Here is a table showing the results of this section. Here is a table showing detailed results for each of the 8 possible cases of random walks restricted to the non-negative axis.

Chapter 3

Operators

In this section we'll make our way towards the larger problem of higher-dimensional labelled walks. The methods will be similar to those in the section on the one-dimensional problem but their complexity will rise substantially. Without the operator notation even the derivation of the functional equation is quite laborious. It's probably worth mentioning that we found the derivation the hard way first before we discovered we can use operators to simplify the process. What once took about a dozen pages now takes a just few. In this section we'll explore the properties of the operators that interest us, showing when the commute, if they have inverses, what their eigenvalues and eigenvectors are and so on.

3.1 Homogeneous Operators

3.1.1 Single 1D Operators

We start off with the observation that an extraction operator E is just a labeling operator L with label $a = 0$. In other words, we can combine the discussion for E and L by combining the operators into one definition. For integer n and a number a let

$$\Gamma_n(a) = 1 + (a-1)x^n[x^n] = 1 + bx^n[x^n]$$

The operators commute because

$$\begin{aligned}\Gamma_n(a_1)\Gamma_m(a_2) &= 1 + b_1 x^n[x^n] + b_2 x^m[x^m] + b_1 b_2 x^n[x^n]x^m[x_m] \\ &= 1 + b_1 x^n[x^n] + b_2 x^m[x^m] + b_1 b_2 \delta_{n,m} x^n[x_n] \\ &= 1 + b_1 x^n[x^n] + b_2 x^m[x^m] + b_1 b_2 x^m[x^m]x^n[x_n] \\ &= \Gamma_m(a_2)\Gamma_n(a_1)\end{aligned}$$

The third term disappears when $n \neq m$. Because of the commutativity we can write a product of such operators in any order we like. We'll usually write them in alphabetical order. Next, we'll find the eigenvalues and eigenfunctions for single operators. Suppose that for some function $u \neq 0$

$$\Gamma_n(a)u = \lambda u$$

where $a \neq 1$ (which would make $\Gamma_n(a) = 1$, the identity operator). We can write $u = u_n x^n + v$ where v has no x^n term. Therefore we find

$$0 = \Gamma_n(a)u - \lambda u = a u_n x^n + v - \lambda u_n x^n - \lambda v = (a-\lambda)u_n x^n + (1-\lambda)v$$

We have two cases

- If $v = 0$ then $u_n \neq 0$ and $\lambda = a$. The eigenfunction is x^n.
- If $v \neq 0$ then $\lambda = 1$ and $u_n = 0$. Here every x^k for $k \neq n$ is an eigenfunction.

This shouldn't be surprising because Γ_n acts as the identity for most of the function except for one term which it multiplies by a. Because of this action of Γ_n a repeated application of the operator leads to a power of a. In other words

$$\Gamma_n(a)^m = 1 + (a^m - 1)x^n[x^n] = \Gamma_n(a^m)$$

When $a \neq 0$ the operator is invertible with inverse

$$\Gamma_n^{-1}(a) = 1 - \left(1 - \frac{1}{a}\right) x^n[x^n]$$

When $a = 0$ the operator has a non-trivial kernel generated by x^n. Furthermore when $a = 0$ the operator is idempotent which is natural given that it just removes the x^n term from the expression it is applied to.

We can think of minimal polynomials too. For example $\Gamma_n(0)$ is non-trivial so its minimal polynomial is not X or $X-1$. However since it's idempotent, its minimal polynomial must divide $X(X-1)$ and since it's neither of the factors it must be the minimal polynomial. On the other hand $\Gamma_n(a)$ for $a \neq 0$ does not have a minimal polynomial because the powers of a are independent from each other.

3.1.2 Compounded 1D Operators

What about a composition of operators? Let $\Gamma = \Gamma_{n_1}(a_1)\dots\Gamma_{n_m}(a_m)$ where the powers n_i are distinct; otherwise treat any repetitions using commutativity and the identity $\Gamma_n(a)^m = \Gamma_n(a^m)$. Using induction with the identity above gives

$$\Gamma = 1 + b_1 x^{n_1}[x^{n_1}] + \cdots + b_m x^{n_m}[x^m]$$

Letting $u = u_{n_1}x^{n_1} + \cdots + u_{n_m}x^{n_m} + v$ in the same way as the single operator case and going through the calculation shows that the eigenvalues of Γ are

- $\lambda = a_i$ for $1 \leq i \leq m$ with eigenfunction x^{n_i}.
- $\lambda = 1$ with eigenfunctions x^m where $m \neq n_i$ for any i.

When one or more of the a_i's is zero we have a non-trivial kernel

$$\ker(\Gamma) = \langle x^m | m = n_i, a_i = 0 \rangle$$

3.1.3 Multidimensional Operators

Multidimensional operators are ones composed of one-dimensional operators (whether single or compounded) and they can also be considered compounded. For this section we'll use superscript notation

$$\Gamma_n^{(k)}(a) = \Gamma^{(k)} = 1 + b_k x_k^n[x_k^n]$$

Operators working on different dimensions commute with each other

$$\Gamma_n^{(i)}\Gamma_m^{(j)} = 1 + b_i x_i^n [x_i^n] + b_j x_j^m [x_j^m] + b_i b_j x_i^n x_j^m [x_i^n x_j^m] = \Gamma_m^{(j)}\Gamma_n^{(i)}.$$

This allows us to write a multidimensional operator in terms of its constituent pieces in whichever order we prefer. We are of course interested in finding eigenvalues and eigenfunctions for such operators. We could either find those with the same method as before or we could find it via induction.

Proposition 1. Let $\Gamma = \Gamma^{(1)} \dots \Gamma^{(d)}$. Then the eigenvalues/eigenfunctions of Γ are the products of eigenvalues/eigenfunctions of the individual $\Gamma^{(i)}$'s. In particular the eigenvalues are products of a's and the corresponding eigenfunctions are products of corresponding powers of x's. The eigenvalue 1 has all the "unseen" monomials as its eigenfunctions. The kernel of Γ consists of multiples of monomials associated to a's that are zero.

Proof. (actually don't know if this will work. The alternative proof is long-winded). We proceed by induction on $d \geq 1$. When $d = 1$ the statement is trivially true. Assuming the statement is true for a collection of $d-1$ operators we consider $\Gamma = \Gamma^{(1)} \dots \Gamma^{(d)} = \Gamma' \Gamma^{(d)}$. □

What is the kernel of Γ?

3.2 Inhomogeneous Operators

We'll call an operator inhomogeneous if it consists of a homogeneous operator Γ and a laurent polynomial (possible infinite) G and and acts in one of the following ways

$$\Gamma S(u) = \Gamma(Su) \quad \text{or} \quad S\Gamma u = S \cdot \Gamma u$$

This kind of operator is important because it features quite a bit in the funcitonal equations of random walk models. We seek to find the eigenpairs for inhomogeneous operators. Our first task is to show that these operators are equivalent in how they work. Suppose that (λ, u) is an eigenpair of ΓS. Let $v = Su$. Then we have

$$S\Gamma S u = S\Gamma S \frac{v}{S} = S\Gamma v = S\lambda u = S\lambda \frac{v}{S} = \lambda v$$

Therefore ΓS and $S\Gamma$ have the same eigenvalues. The eigenvectors of ΓS as the same as those for $S\Gamma$ by divided by S. Now if we pick $\lambda = 0$ to find the kernels of the operators we find

$$S\ker(\Gamma S) = \ker(S\Gamma) = \ker(\Gamma)$$

and thus

$$\ker(\Gamma S) = \overline{S}\ker(\Gamma)$$

The functions in this kernel are specifically those of the kernel of Γ but multiplied by $1/S$.

3.2.1 Single 1D Operators

Let G and u be given as

$$S = \sum_k S_k x^k, \quad u = \sum_k u_k x^k$$

where u is an eigenfunction of ΓS. We have

$$\begin{aligned} 0 &= (\Gamma S - \lambda I)u \\ &= \sum_{k \neq n} x^k \left(\sum_i S_i u_{k-i} - \lambda u_i \right) + x^n \left(a \sum_i S_i u_{n-i} - \lambda u_n \right) \end{aligned}$$

which leads to the system of equations

$$\begin{aligned} \lambda u_n &= a \sum_i S_i u_{n-i} \\ \lambda u_k &= \sum_i S_i u_{k-i}, \quad \text{if } k \neq n \end{aligned}$$

The analysis is non-trivial even for a simple function S like $S = x$. We have

$$\begin{aligned} \lambda u_n &= a u_{n-1} \\ \lambda u_k &= u_{k-1}, \quad \text{if } k \neq n \end{aligned}$$

We have a doubly-infinite recursion so we can pick an arbitrary initial condition. If $u_n = 0$ the rest of the terms will be zero as well so we must have $u_n \neq 0$. Set

$u_n = 1$. Then via induction we can find

$$u_{n+k} = 1/\lambda^k, \quad \text{for } k \geq 0$$
$$u_{n-k} = \lambda^k/a^k, \quad \text{for } k \geq 1$$

Computing $u(x)$ formally leads to

$$\begin{aligned} u(x) &= \sum_{k=1}^{\infty} u_{n-k}x^{n-k} + \sum_{k=0}^{\infty} u_{n+k}x^{n+k} \\ &= x^n \sum_{k=1}^{\infty} \left(\frac{\lambda}{ax}\right)^k + x^n \sum_{k=0}^{\infty} \left(\frac{x}{\lambda}\right)^k \\ &= \frac{x^{n-1}x}{ax-\lambda} - \frac{x^n\lambda}{x-\lambda} \\ &= \lambda x^n \left(\frac{1}{\lambda - x} - \frac{1}{\lambda - ax}\right) \\ &= -\frac{\lambda b x^{n+1}}{(\lambda - x)(\lambda - ax)} \end{aligned}$$

We could also try using the other definition. Let $u = u_n x^n + v$ where u is now an eigenfunction of $S\Gamma$. Let $S = S_0 + S'$. Then we have

$$\begin{aligned} S\Gamma u &= (S_0 + S')(au_n x^n + v) = \lambda(u_n x^n + v) \\ 0 &= (au_n S_0 - \lambda u_n + \sum_{k \neq n} S_k u_{n-k})x^n + au_n S' x^n + S'v - \lambda v \end{aligned}$$

We find that for example

$$(\lambda - aS_0)u_n = \sum_{k \neq n} S_k u_{n-k}$$

which doesn't seem to help much.

3.3 Random Walks Operator

In this section we explore the operator $\mathscr{L} = 1 - t\Gamma S$ for a step Laurent polynomial S and some random walk operator Γ. We haven't made it rigorous yet but $\mathscr{L}$ is

invertible with inverse

$$\overset{-1}{\mathscr{L}} = \sum_{k=0}^{\infty} t^k (\Gamma S)^k$$

where $(\Gamma S)^k$ is an iteration of the inhomogeneous operator with $(\Gamma S)^0 = I$. However this inverse is not easy to use as the iteration become quite complicated due to the nature of the random walks. Let (u, λ) be an eigenpair for $\mathscr{L}$.Then we calculate

$$\begin{aligned}(\mathscr{L} - \lambda I)u &= u - \lambda u - t\Gamma S u \\ t\Gamma S u &= (1-\lambda)u = \mu u\end{aligned}$$

If $\lambda = 1$ we seek to find the kernel of ΓS which we did in a previous section. The eigenfunctions of $\mathscr{L}$ for eigenvalue $\lambda = 1$ are the functions $u \in \overline{S}\ker\Gamma$.

Next let u be given by a finite order Laurent series

$$u = u_n t^n + u_{n+1} t^{n+1} + \dots$$

where $u_n \neq 0$. Then we can re-write the functional equation as

$$t^{-n}(1 - t\Gamma S)u = \lambda t^{-n} u.$$

Letting $t \to 0$ in this equation yields

$$(\lambda - 1)u_n = -\lim_{t\to 0} t^{1-n}\Gamma S u = -\Gamma S \lim_{t\to 0}(t^{1-n}u) = 0.$$

Since $u_n \neq 0$ this implies that $\lambda = 1$ and again we see that $u \in \overline{S}\ker\Gamma$. Since S is a Laurent polynomial, the resulting functions will indeed be finite order Laurent series. For the remainder of the analysis we assume that $\lambda \neq 1$. This excludes finite order Laurent series so u is necessarily given by an infinite order Laurent series.

Let's investigate how how ∂_t interacts with $\mathscr{L}$.

$$\begin{aligned}\partial_t \mathscr{L} &= \partial_t - \partial_t(t\Gamma S) = \partial_t - \Gamma S - t\Gamma S\partial_t = \mathscr{L}\partial_t - \Gamma S \\ \partial_t^2 \mathscr{L} &= \mathscr{L}\partial_t^2 - 2\partial_t \Gamma S \\ \partial_t^n \mathscr{L} &= \mathscr{L}\partial_t^n - n\partial_t^{n-1}\Gamma S\end{aligned}$$

for integer n. Another way to write these is with bracket notation

$$[\mathscr{L}, \partial_t^n] = \mathscr{L}\,\partial_t^n - \partial_t^n\,\mathscr{L} = n\partial_t^{n-1}\Gamma S$$

A question that pops up is how does u relate to its derivatives.

$$\partial_t\,\mathscr{L}\,u = \mathscr{L}\,u' - \Gamma S u = \lambda\,u'$$

However

$$\Gamma S u = \frac{\mu u}{t}$$

from before so

$$\mathscr{L}\,u' = \lambda\,u' + \frac{\mu u}{t}$$

Chapter 4

Unlabeled Walks

4.1 Interactions with Operators

4.2 Orbit Sums Notation

4.3 Infinite Group Formulations

Chapter 5

Higher-dimensional labelled walks

5.1 Using Operators to Derive the Functional Equation

As in the 1D case we'll take the generating function Q, multiply it by tSQ to obtain all the generating function for all walks counted in Q and taken a step further by the step set S. Not all of these are valid so we need to remove the ones that are not. We can apply operator E_1 to retain only those walks that are x_1-valid. We can apply E_2 on the remaining walks to retain only those walks that are x_2-valid as well. We continue like this until E_d is applied and we only have valid walks remaining. Then we can take care of the labelling by applying operator L_1. This takes all the walks counted in tSQ that have ended on the $x_1 = 0$ hyperplane, removes then and re-adds their labelled versions. We can continue applying operators L_i until L_d. In the end we've retained all the valid walks that are also properly re-labelled as they have interacted with the hyperplanes $x_i = 0$. This quatity is $tLE(SQ)$. The only walk from Q that is not present is 1, the empty walk. Therefore we have our functional equation

$$Q = 1 + tLE(SQ)$$

which we can re-write as

$$(1 - tLES)Q = 1$$

We can apply the operator M on this to obtain

$$M(1) = \sum_{I \subseteq [d]} c_I = \overline{a}_{[d]}$$

and

$$(M - tES)Q = \overline{a}_{[d]}$$

This equation will contain many less terms. To see this we take a look at the $L_i E_i$ operator

$$L_i E_i = 1 + b_i[x_i^0] - \overline{x_i}[\overline{x_i}]$$

which increases the number of terms in the equation by a factor of 3. That is we would have 3^d terms relating to SQ by the end of applying LE. Not to mention that each extraction from SQ introduces at most 3 terms. On the other hand once we apply the operator M we introduce 2^d terms relating to Q but then since we only have E applied on SQ we'd only have at most 2^d terms from there. For the case of walks of step size at most 1 and labelling the hyperplane interactions, these terms turn out to be mostly the same. We have a lot fewer terms overall in the latter representation. In this case we can write out the functional equation explicitly. Firstly,

$$MQ = (1 - c_1[x_1^0]) \dots (1 - c_d[x_d^0])Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I [x_I^{\mathbb{O}}]Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{O}}$$

and

$$\begin{aligned} ESQ &= (1 - \overline{x}_1[\overline{x}_1]) \dots (1 - \overline{x}_d[\overline{x_d}])(SQ) \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [x_I^{-\mathbb{I}}](SQ) \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}} \end{aligned}$$

Therefore the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = \overline{a_{[d]}}$$

or if we let $K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$ we have

$$\sum_{I\subseteq[d]} (-1)^{|I|} K_I Q_I^{\mathbb{O}} = \overline{a_{[d]}}.$$

5.2 Generalizations to the functional equation

Using the operator notation we can handle generalizations fairly easily. First we can consider a more general step set. Let n_i be non-negative integers for $1 \leq i \leq d$. Then let S be a step set for which no step backtracks more than n_i units in the negative x_i direction. The positive direction is bounded. Then we can write the extraction operators as

$$E_i = 1 - \overline{x}_i[\overline{x}_i] - \ldots - \overline{x}_i^{n_i}[\overline{x}_i^{n_i}]$$

The reason why this works is that this operator must remove all walks that are not x_i-valid and this operator removes all the walks once they step beyond the $x_i = 0$ plane. Of course we could easily generalize to exclude certain step sizes. For example, if the walks have a certain step length, they might be able to jump over wide boundaries (this might be useful in some models). Another example is where we might want to look at all quarter plane walks that don't step at a certain point with coordinates (m,n). Then we can write an operator

$$E(m,n) = 1 - x_1^m x_2^n [x_1^m x_2^n]$$

that we'd apply on SQ to remove those walks that end on (m,n).

Next we take care of labelling. We might want to label the interactions not just with the boundaries but also with specific hyperplanes. For example, if we want to label the boundary interactions and also the interactions with the closest 2 hyperplanes near the $x_i = 0$ hyperplane in the direction of the x_i axis we'd write

$$L_i = 1 + b_{i,0}[x_i^0] + b_{i,1}x_i^1[x_i^1] + b_{i,2}x_i^2[x_i^2]$$

Of course we can also easily label the interactions with specific points. If we want

to label the interactions with point (m,n) with a label $a_{m,n}$ we write

$$L(m,n) = 1 + b_{m,n} x_1^m x_2^n [x_1^m x_2^n]$$

It might be interesting to consider the interactions of quarter plane walks with the line $x_2 = (q/r)x_1$ for some coprime integers q, r. For this we need the label

$$L = 1 + b \sum_{k=0}^{\infty} x_1^{qk} x_2^{rk} [x_1^{qk} x_2^{rk}]$$

This is going to get a little crazy right away so I'm sorry. I just thought of the most general thing I could. All these operators can be thought of as belonging to a family of more generalized operators

$$\Gamma_k(x_1, \ldots, x_d) = \sum_{\substack{I \subseteq [d] \\ |I|=k}} \sum_{J \in \mathbb{Z}^k} \gamma_{I,J} x_I^J [x_I^J]$$

For example the last one can be written as

$$L = 1 + b\Gamma_1(x_1^q x_2^r)$$

for $\gamma_k = 1$ and the first one in the section as

$$E_i = 1 - \Gamma_1(\overline{x}_1)$$

with the sequence $\gamma_k = 1$ for $0 \leq k \leq n_i$. It's more of a thought right now, not something we'll really look into. It suffices to say that all of these operators will commute because there is no mixing of terms in the extractions they perform. We may ask questions about inverses and kernels. The operators E remove terms from the function they are applied on so they are irreversible and so have a non-zero kernel. The labelling operators change terms by a constant so it is enough to remove those constants. For example for the first labelling operator

$$M_i = 1 - c_{i,0}[x_i^0] - c_{i,1} x_i^1 [x_i^1] - c_{i,2} x_i^2 [x_i^2]$$

and then

$$M(m,n) = 1 - c_{m,n} x_1^m x_2^n [x_1^m x_2^n]$$

This is because $-c = 1/a-$ so the operator removes the old term and replaces it with one multiplied by $1/a$. This discussion allows us to still write the functional equation as

$$(1 - tLES)Q = 1$$

and upon applying M we have

$$(M - tES)Q = M(1)$$

where $M(1)$ will depend only on the labelling of contacts with the hyperplanes $x_i = 0$ and not any other points. We could write out explicit equations for some operators.

5.3 Coefficient Extraction

We can perform coefficient extraction like in the 1-dimensional case. The operator $[x_i^k]$ will commute with all M_j, E_j with $j \neq i$ so that we only need to consider the action on the operators M_i and E_i. We've already done so in the 1-dimensional case. The zeroth power extraction yields

$$[x_i^0]M_i = [x_i^0] - [x_i^0]c_i[x_i^0] = [x_i^0] - c_i[x_i^0] = \overline{a_i}[x_i^0]$$
$$[x_i^0]E_i = [x_i^0] - [x_i^0]\overline{x_i}[\overline{x_i}] = [x_i^0]$$

and the k-th power extraction for $k \geq 1$ yields

$$[x_i^k]M_i = [x_i^k] - [x_i^k]c_i[x_i^0] = [x_i^k]$$
$$[x_i^k]E_i = [x_i^k] - [x_i^k]\overline{x_i}[\overline{x_i}] = [x_i^k]$$

So, for $k \geq 0$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture while for $k = 0$ it removes E_i but leaves a residual $\overline{a_i}$ behind for M_i. Therefore if we extract

$[x_I^{\mathbb{O}}]$ from the functional equation we have

$$\begin{aligned}[x_I^{\mathbb{O}}]a_{[d]} = \overline{a}_{[d]} &= [x_I^{\mathbb{O}}](M_{[d]} - tE_{[d]}S)Q \\ &= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{O}}] - tE_{[d]\setminus I}[x_I^{\mathbb{O}}]S\right)Q \\ &= [x_I^{\mathbb{O}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q\end{aligned}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{O}}](M_{[d]\setminus I} - a_I tE_{[d]\setminus I}S)Q$$

Written in another way this becomes

$$\overline{a}_I = [x_{[d]\setminus I}^{\mathbb{O}}](M_I - a_{[d]\setminus I}tE_I S)Q$$

and we can use whichever notation is more useful at the time. The latter notation underscores the idea that when we extract coefficients for walks that end at some hyperplane intersection we are having a reduced version of the functional equation at the cost of an added a term. When we extract a general term we can use the operator $[x_I^{\mathbb{O}} x_J^P]$ where P is a set of powers greater than 0 and $I \cap J = \emptyset$, $|P| \geq 1$ and $I \cup J = K$. We find

$$\begin{aligned}[x_I^{\mathbb{O}} x_J^P]a_{[d]} = 0 &= [x_I^{\mathbb{O}} x_J^P](M_{[d]} - tE_{[d]}S)Q \\ &= [x_I^{\mathbb{O}} x_J^P](\overline{a}_I M_{[d]\setminus K} - tE_{[d]\setminus K}S)Q\end{aligned}$$

which we can write as

$$[x_I^{\mathbb{O}} x_J^P](M_{[d]\setminus K} - a_I tE_{[d]\setminus K}S)Q = 0$$

If I is empty then

$$[x_J^P](M_{[d]\setminus J} - tE_{[d]\setminus J}S)Q = 0$$

The problem with a lot of things in this section is that they are not really practical beyond providing a shortcut for some of the calculations so we don't have to do them by hand every time. Also they provide a broader idea of having multiple kernels in one problem. Perhaps things will be clearer when we specialize to the

2-dimensional problem.

5.4 Further functional equations

We can try going a little further in the coefficient extraction to come up with an equation that looks like the original functional equation. We know that

$$[x_I^{\mathbb{O}}]SQ = S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I$$

where R_I is a collection of $S_I^{-P} Q_I^P$ terms that involves sets P whose terms are not all zero. We can use this inside the functional equation

$$\begin{aligned}
\overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{O}}]Q - a_I t E_{[d]\setminus I}[x_I^{\mathbb{O}}](SQ) \\
&= M_{[d]\setminus I} Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I}(S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I) \\
&= (M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S_I^{\mathbb{O}}) Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I} R_I
\end{aligned}$$

so we can re-write the functional equation as

$$(M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S_I^{\mathbb{O}}) Q_I^{\mathbb{O}} = a_I(\overline{a}_{[d]} + t R_I')$$

where $R_I' = E_{[d]\setminus I} R_I$. We can calculate the following terms

$$M_{[d]\setminus I} Q_I^{\mathbb{O}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} c_J Q_{I,J}^{\mathbb{O},\mathbb{O}}$$

and

$$a_I t E_{[d]\setminus I} S_I^{\mathbb{O}} Q_I^{\mathbb{O}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} a_I t \bar{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}} Q_{I,J}^{\mathbb{O},\mathbb{O}}$$

so the functional equation can be written as

$$\sum_{J \subseteq [d]\setminus I} (-1)^{|J|} (c_J - a_I t \bar{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}}) Q_{I,J}^{\mathbb{O},\mathbb{O}} = a_I(\overline{a}_{[d]} + t R_I')$$

This suggests the notation

$$K_J(I) = c_J - a_I t \bar{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}}$$

where when $I = \emptyset$, $K_J(I) = K_J$ from before. This allows us to write the 2^d functional equations in the simplified form

$$\sum_{J \subseteq [d] \setminus I} (-1)^{|J|} K_J(I) Q_{I,J}^{\mathbb{O},\mathbb{O}} = \overline{a}_{[d]\setminus I} + a_I t R'_I$$

where the R'_I term involves only higher order coefficient extractions from Q, that is not all the powers are zero. Now, to calculate the R'_I terms we first determine what R_I is. It is a sum of terms $S_I^{-P} Q_I^P$ for various nonnegative P with size $|I|$ whose elements are not all 0. To calculate R'_I we apply the operator $E_{[d]\setminus I}$ on R_I. We can try calculating some of these quantities. A key observation is that the terms Q_I^P are power series in the remaining variables so they don't have non-negative coefficients. Let $sq = S_I^{-P} Q_I^P$. Then

$$\begin{aligned} E_j sq &= sq - \overline{x}_j (sq)_j^{-1} \\ &= sq - \overline{x}_j (s_j^{-1} q_j^0 + s_j^0 q_j^{-1} + s_j^1 q_j^{-2}) \\ &= sq - \overline{x}_j s_j^{-1} q_j^0 \end{aligned}$$

Hence

$$E_K sq = \sum_{J \subseteq K} (-1)^{|J|} \overline{x}_J s_J^{-\mathbb{I}} q_J^{\mathbb{O}}$$

Finally we can write

$$\begin{aligned} R'_I &= E_{[d]\setminus I} R_I \\ &= \sum_{P > \mathbb{O}} E_{[d]\setminus I} S_I^{-P} Q_I^P \\ &= \sum_{P > \mathbb{O}} \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} \\ &= \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{O}} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} \end{aligned}$$

This is not a very nice formula to use but it should do for now.

5.5 Using the formal inverse for iteration

In a previous subsection we obtained the equation

$$(1 - tLES)Q = 1$$

Let $D = LE$. Then we can think of the formal inverse of this operator against some norm $||tDS||$ or even formally. Formally

$$(1 - tDS)^{-1} = \frac{1}{1 - tDS} = \sum_{k=0}^{\infty} t^k (DS)^k$$

so we can write the solution for Q as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k (1)$$

where $(DS)^k(1) = DS(\ldots(DS(1))\ldots)$. This allows us to compute the series expansion for Q directly using an iteration. The next coefficient is the previous one with DS applied to it. It can be done fairly efficiently on a computer. Interestingly enough this is just a formalization of the brute force method of determining the functional equation - by taking a previous set of walks, extending them by 1 step, removing the ones that fall outside of the quarter plane and finally labelling the ones that have interacted with the hyperplanes $x_i = 0$ in the process. We could attempt to write a closed form for the operator application but it becomes messy fairly fast, even for one dimension.

Chapter 6

Other Flavoured Walks

6.1 Walks with other constraints

Random walk models with additional constraints are handled naturally via the operator notation and herein lies the full power of the framework. The operator notation makes for a quick and easy derivation of master functional equations for most such models. Using extraction operators we can constrain random walks within a finite region, semi-infinite strip, a semi-infinite sector or any other discrete shape (with diminishing returns for more complicated shapes of course). Furthermore using labeling operators we can easily track the behavior of random walks as they cross or interact with boundaries or interact with objects. The exposition can be combined to talk about a lebeling operator Γ and then setting some of the labels equal to zero to make it an extraction operator.

6.1.1 Regional Constraints

To find the generating function for random walks confined within a region D all we need to do is take all walks and remove those that venture outside of D. This cannot be done with the final form of the each random walk because it doesn't have memory. However we can do this as an iterative process. We grow random walks from within the region D and once a walk ventures outside of D we snip it using an operator E. By growing all random walks one step at a time and removing all those that venture outside of D we will be left with only those that remain within at all

times. It is not difficult to find what E should be. Suppose we have the boundary ∂D on the integer lattice. We are allowed to have walks step on the boundary but not beyond it. Given a specific step set S we can start with a generating function B for step 0 walks originating on the boundary (basically the coordinates of ∂B) and iterate by one step, that is, compute the product SB. We only keep the terms in this product that are outside of the boundary that is $B_0 = SB \cap D^c$ in terms of monomials. Then we can write the extraction operator as simple coefficient extraction featuring the terms in B_0. That is

$$E_0 = 1 - \sum_{e \in B_0} e[e]$$

This is the most accurate representation of what we want E to be. However it's useful to simplify the discussion by taking advantage of a trivial fact: the effect of E_0 doesn't change if we add some more terms from the outside of the region that can be reached from the boundary. As an illustration, for quarter-plane walks with small steps E_0 is composed of the terms $e[e]$ where $e = x_1^{-1}x_2^i$ or $e = x_1^i x_2^{-1}$ for $i \geq -1$. However it's much useful to add in those terms with powers $(-1, i)$ or $(i, -1$ for $i < -1$. In this case the new operator is

$$\begin{aligned}
E &= 1 + \bar{x}_1\bar{x}_2[\bar{x}_1\bar{x}_2] \sum_{i=-\infty}^{\infty} (\bar{x}_1 x_2^i[\bar{x}_1 x_2^i] + x_1^i\bar{x}_2[x_1^i\bar{x}_2]) = \\
&= 1 - \bar{x}_1[\bar{x}_1] \sum_{i=-\infty}^{\infty} x_2^i[x_2^i] + \bar{x}_2[\bar{x}_2] \sum_{i=-\infty}^{\infty} x_1^i[x_1^i] + \bar{x}_1\bar{x}_2[\bar{x}_1\bar{x}_2] \\
&= 1 - \bar{x}_1[\bar{x}_1] + \bar{x}_2[\bar{x}_2] + \bar{x}_1\bar{x}_2[\bar{x}_1\bar{x}_2] \\
&= (1 - \bar{x}_1[\bar{x}_1])(1 - \bar{x}_2[\bar{x}_2]). && = E_1E_2
\end{aligned}$$

This operator is much easier to work with. If we want to enumerate walks with small steps in the upper half-plane we should use $E = E_2$. If we have to enumerate walks with small steps within a semi-infinite strip with coordinates $x_1 \geq 0$ and $0 \leq x_2 \leq n$ we can use

$$E = (1 - \bar{x}_1[\bar{x}_1])(1 - \bar{x}_2[\bar{x}_2] - x_2^{n+1}[x_2^{n+1}])$$

For walks within a rectangle of sides m,n in the positive quarter plane

$$E = (1-\bar{x}_2[\bar{x}_2]-x_2^{m+1}[x_2^{m+1}])(1-\bar{x}_2[\bar{x}_2]-x_2^{n+1}[x_2^{n+1}])$$

For walks on the quarter plane within a rectangle of sides $2m+1, 2n+1$ centered at the origin, we have

$$E = (1-\bar{x}_2^{m+1}[\bar{x}_2^{m+1}]-x_2^{m+1}[x_2^{m+1}])(1-\bar{x}_2^{n+1}[\bar{x}_2^{n+1}]-x_2^{n+1}[x_2^{n+1}])$$

6.1.2 Labeling Special Locations

The analysis works in the same way as the one for the regional constraints. If we have different regions with different labels we can just split the work into independent cases. Let D be the region which we wish to label with a boundary ∂D on the integer lattice. Let B be the boundary Laurent polynomial. Then our labeling operator is simply

$$L_0 = 1 + (a-1)\sum_{e\in B} e[e]$$

As was the case before we want to add in some more terms to L_0 to make it easier to use. For example - in the case of quarter-plane walks with small steps we want to label walks visiting the axes. To label those that touch the x_1-axis (second coordinate is 0) we set the region to be the upper quarter plane with boundary the x_1-axis. Therefore we let

$$L_2 = 1 + b_2\sum_{i=-\infty}^{\infty} x_1^i x_2^0[x_1^i x_2^0] = 1 + b_2 x_2^0[x_2^0]\sum_{i=-\infty}^{\infty} x_1^i[x_1^i] = 1 + b_2 x_2^0[x_2^0]$$

Similarly

$$L_1 = 1 + b_1 x_1^0[x_1^0]$$

and finally

$$\begin{aligned} L = L_0 = L_1 L_2 &= (1 + b_1 x_1^0[x_1^0])(1 + b_2 x_2^0[x_2^0]) \\ &= 1 + b_1 x_1^0[x_1^0] + b_2 x_2^0[x_2^0] + b_1 b_2 x_1^0 x_2^0[x_1^0 x_2^0] \end{aligned}$$

6.2 Self-avoiding walks

This one is a bit more ad hoc because we have to keep a history of where the random walks go. Furthermore the operators involved require a sort of limiting argument. For the first n steps we "only" require an operator covering the square region of n^2 lattice points. The overall operator is defined through a limiting process of applying the full operator (or truncated version) on a finite Laurent polynomial. Let

$$\Gamma_{i,j} = 1 + b_{i,j} x_1^i x_2^j [x_1^i x_2^j]$$

where as usual $b_{i,j} = a_{i,j} - 1$. This operator takes a generating function and labels the monomial $x_1^i x_2^j$. We'll be using a lot of these together so we let

$$\Gamma_n = \prod_{-n \le i,j \le n} \Gamma_{i,j}$$

To find a functional equation for SAWs on the plane we perform an iteration. Let Q be the generating function for walks with history, that is they have a label for each coordinate they pass through and let $Q^{\le n}$ be the truncated version of this generating function. It's given by

$$Q^{\le n} = \sum_{k=0}^{n} Q_k t^k$$

where the initial condition is $Q_0 = a_{0,0}$. Starting from Q_n we can find Q_{n+1} by iterating with the step set S, labeling with Γ_{n+1} as follows

$$Q_{n+1} = \Gamma_{n+1} S Q_n$$

Using a computer we can easily iterate this to get more and more steps. We can also get $Q^{\leq n+1}$ from $Q^{\leq n}$ since

$$\begin{aligned} t\Gamma_{n+1}SQ^{\leq n} &= \sum_{k=0}^{n} \Gamma_{n+1}SQ_k t^{k+1} \\ &= \sum_{k=0}^{n} \Gamma_{k+1}SQ_k t^{k+1} \\ &= \sum_{k=0}^{n} Q_{k+1} t^{k+1} \\ &= Q^{\leq n+1} - Q_0 \end{aligned}$$

or in other words

$$Q^{\leq n+1} = a_{0,0} + t\Gamma_{n+1}SQ^{\leq n}$$

In the interest of finding a functional equation we would like to take the limiit of this as $n \to \infty$. For this we define Γ to be the operator labeling all monomials $x_1^i x_2^j$ and acting in the following way

$$\Gamma x^i y^j = \Gamma_{\max(|i|,|j|)} x^i y^j$$

and extended by linearity. The limiting process is

$$Q = a_{0,0} + t\Gamma SQ$$

We finally have the functional equation for random walks with full history

$$(1 - t\Gamma S)Q = a_{0,0}.$$

From this we can obtain the generating function W for the SAWs by removing all walks with power of label $a_{i,j}$ greater than 1. For this we need a removal operator.

Let

$$R_{i,j} = a^0_{i,j}[a^0_{i,j}] + a_{i,j}[a_{i,j}]$$
$$R_n = \prod_{-n \le i,j \le n} R_{i,j}$$
$$R = \lim_{n\to\infty} R_n$$

where the limit is understood as in the example of Γ above. The solution for Q can also be restated as

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k (a_{0,0}) = \sum_{k=0}^{\infty} t^k (\Gamma_k S)^k (a_{0,0})$$

We can either redo the analysis applying R at every iteration or we can apply R to Q and get

$$W = RQ = \sum_{k=0}^{\infty} t^k R(\Gamma S)^k (a_{0,0}) = \sum_{k=0}^{\infty} t^k (R\Gamma S)^k (a_{0,0})$$

We can write this as a functional equation

$$\mathscr{L} W = (1 - t\Sigma)W = (1 - tR\Gamma S)W = a_{0,0}$$

6.2.1 Eigenpairs

We can try finding eigenvalues and eigenfunctions of $\mathscr{L}$. Let $\mathscr{L} u = \lambda u$. Writing it out leaves us with

$$t\Sigma u = (1 - \lambda)u$$

So if $\lambda = 1$, $u \in \ker(\Sigma)$. That is $R\Gamma S u = 0$. In other words $\Gamma S u$ has only terms that involve powers greater than 1 of $a_{i,j}$ for various i, j. Suppose that

$$\Gamma S u = a^2_{i,j} f(x_1, x_2)$$

Then

$$u = \frac{a^2_{i,j}}{S} \Gamma^{-1}(f)$$

So we here we can let

$$f = a_{k,l}^m x_1^r x_2^s t^n$$

and we find that

$$u = \frac{a_{i,j}^2 a_{k,l}^m}{a_{r,s} S} x_1^r x_2^s t^n$$

Thus the kernel of Σ is generated by these functions for all coefficients $i, j, k, l, r, s, n \in \mathbb{Z}$ and $m \geq 0$. This may perhaps prove useful later on. Now, if we look at eigenvalues $\lambda \neq 1$ we have to solve the equation

$$t\Sigma u = (1-\lambda)u$$

Because of the t, if u is a finite order laurent series in t with first non-zero coeficient u_n. We find that since Σ does not affect the t monomials

$$\Sigma u_n = \mu u_{n-1} = 0$$

and this forces all u_k coefficients to be zero. Therefore Σ has no other eigenvalues except 1 among the finite order power series in t.

6.2.2 Generalizations

The discussion lends itself nicely to generalization where now we have d variables and operators $\Gamma_{i_1,\ldots,l_d}$, $R_{i_1,\ldots,i_d}$, etc defined analogously. The eigenpair discussion works out in the same way with eigenfunctions for $\lambda = 1$

$$u = \frac{a_{i_1,\ldots,i_d}^2 a_{j_1,\ldots,j_d}^m}{a_{k_1,\ldots,k_d} S} x_1^{r_1} \ldots x_d^{r_d} t^n$$

We can also constrain the SAWs on a part of the plane with an extraction operator E and label special locations with an operator L. The functional equation is

$$(1 - tLE\Sigma)W = a_{0,\ldots,0}$$

for arbitrary number of dimensions, with region constraints and labeling of special locations.

6.3 Other exotic walks

Here we'll have a small exploration of other exotic random walk models.

6.4 Reformulations of Problems

6.4.1 Using labeling operators instead of extraction operators

An initially promising idea is to use a labelling operator instead of an extraction operator with the goal of removing those labeled walks at a later stage. The added benefit is that we would be working with invertible operators throughout the majority of the analysis. The downside is that now the walks will not be constrained and we may get further issues. So, instead of

6.4.2 Considering a symmetrical super-problem

An extension of the previous idea is to symmetrize the problem by also labeling parts on the other side of the origin. Instead of just labeling walks on the axes and on lines where one coordinate is -1 we can also label walks on lines with one coordinate $+1$. This leads to a symmetrical arrangement of labeling operations on both sides of the origin. We can skip the labeling of the axes if we don't wish to investigate wall interactions.

Chapter 7

Hexagonal Walks

7.1 Introduction

We'll tackle the problem of random walks on a honey-comb lattice. This is an example of a problem set on a different topology. It's our first problem dealing with a random walk model where the direction in which you go depends on where you are on the lattice. It can be interpreted as a random walk model on the plane where we have two distinct step polynomials depending on where we are. Because of this the flavour of the problem will be quite different. However it shall serve as a good example of the use of operators. We start by embedding the honeycomb lattice in $\mathbb{Z}^2$ where the first hexagon has coordinates

$$\{(0,0),(1,1),(1,2),(0,0),(-1,2),(-1,1)\}$$

and the rest of the lattice is obtained by tiling the plane with this hexagon. As before we let Q be the generating function for the random walks on the lattice where x, y keep track of the endpoint of a walk and t keeps track of its length. Of course, off the lattice the terms will have zero coefficients. Out of thin air we create two operators C_1 and C_2, acting on Q, which pick out all the bottom and top corners of the hexagon tiles respectively. By picking out we mean that C_1 results in the partial generating function corresponding the the coordinates given in the two situations. The condition is that $C_1 + C_2 = 1$ the identity. These two locations give two distinct

possibilities for step sets so we'll give them their own step polynomials. Let's call their respective step polynomials S_1 and S_2 given by

$$S_1 = (x+\overline{x})y+\overline{y}, \quad S_2 = (x+\overline{x})\overline{y}+y$$

Now we can continue to create a functional equation for the hexagonal model. We perform an iteration. We start with the empty walk with generating function $Q = 1$. Then for every iterate of Q we create the next iterate by taking the walks enumerated in Q and extending them by one step. Since we have two situations we need to extend by different steps in each case. That is we start with Q, extract the S_1-centric walks with C_1Q and then extend them by multiplying with S_1. The result is that the new Q is given by

$$Q = 1+t(S_1C_1+S_2C_2)Q = 1+t\Sigma Q.$$

We can also write this in the operator format as

$$(1-t\Sigma)Q = 1$$

We may wish to restrict the random walks to the quarter plane or in some other way. To do this we need to also use an extraction operator E that removes those walks that stray out of the boundaries. Since each step is of size at most 1 we only need to remove those walks that step 1 unit outside of the boundaries and E is then easy to write. The iteration and functional equation are given by

$$Q = 1+tE(S_1C_1+S_2C_2)Q = 1+tE\Sigma Q$$

$$(1-tE\Sigma)Q = 1$$

Of course we can also label the walks that interact with specific locations. We repeate the work with the extra operator L. In general we can use an operator Γ to encapsulate everything extra that we do. It remains to find the actual operators C_1 and C_2.

7.2 Operators

Before we find the operators we need to find which points they need to act on. We can find all the lower vertices by starting with the hexagon with lower vertex $(0,0)$ and translating it along the lattice. It's not difficult to see that the lower vertices of these hexagons are those with coordinates (m,n) where

- $m \equiv 0 \pmod 2, n \equiv 0 \pmod 4$
- $m \equiv 1 \pmod 2, n \equiv 2 \pmod 4$

The upper vertices of these hexagons are the above coordinates translated by $(0,3)$ in the vertical direction. Those are the coordinates (m,n) where

- $m \equiv 0 \pmod 2, n \equiv 3 \pmod 4$
- $m \equiv 1 \pmod 2, n \equiv 1 \pmod 4$

This allows us to write the operators as

$$C_1 = \sum_{m,n} x^{2m} y^{4n} [x^{2m} y^{4n}] + \sum_{m,n} x^{2m+1} y^{4n+2} [x^{2m+1} y^{4n+2}]$$
$$C_2 = \sum_{m,n} x^{2m+1} y^{4n+1} [x^{2m+1} y^{4n+1}] + \sum_{m,n} x^{2m} y^{4n+3} [x^{2m} y^{4n+3}]$$

We can write these operators more succinctly by using the following conventions

$$d_{j,k}(z) = \sum_m z^{jm+k} [z^{jm+k}]$$
$$D_{ij} = d_{2,i}(x) d_{4,j}(y)$$
$$D = D_{00}$$

Note that

$$D_{ij} = x^i y^j D \bar{x}^i \bar{y}^j$$

This allows us to write the operators C_1 and C_2 succinctly as

$$C_1 = D_{00} + D_{12} = D + xy^2 D\overline{xy}^2$$
$$C_2 = D_{03} + D_{11} = y^3 D\overline{y}^3 + xyD\overline{xy}$$

The functional equation now reads

$$(1 - t(S_1(D_{00} + D_{12}) + S_2(D_{03} + D_{11}))Q = 1$$

and analogously we can append an operator Γ to to the expression.

7.3 Group Structure

There are nicer ways to express these extractions without performing the extractions themselves. Let k be a positive integer and $w_k = e^{2\pi i/k}$. Then it is a somewhat well-known result that we can extract the coefficient $0 \pmod k$ of $f(x)$ with the following summation

$$d_{k,0}(x)f(x) = \frac{1}{k}\sum_{j=0}^{k-1} f(w_k^j x).$$

This establishes the concrete operator representations

$$d_{2,0}(x)f(x) = \frac{1}{2}(f(x) + f(-x))$$
$$d_{4,0}(y)f(y) = \frac{1}{4}(f(y) + f(iy) + f(-y) + f(-iy))$$

Using this definition in the functional equation would lead to a functional equation relating $Q(x,y)$ to functions Q evaluated at different arguments like $(-x,y)$, (x,iy), etc. This allows us to explore the group structure of the functional equation. To this end let $g : x \mapsto -x$ and $h : y \mapsto iy$ be the two commuting operations and the group of symmetries be

$$G \cong \langle g \rangle \times \langle h \rangle \cong \langle -1 \rangle \times \langle i \rangle$$

It remains to see how these group actions affect the various quantities in the problem. First we deal with S_1 and S_2 and then with the operators D_{jk}. This will allow

us to find how the actions affect Σ and the functional equation as a whole. First we have the actions of g

$$gS_1 = S_1^* = -(x+\bar{x})y+\bar{y}$$
$$gS_2 = S_2^* = -(x+\bar{x})\bar{y}+y$$

Then the actions of h

$$hS_1 = i(x+\bar{x})y - i\bar{y} = -igS_1 = -iS_1^*$$
$$hS_2 = -i(x+\bar{x})\bar{y} + iy = igS_2 = iS_2^*$$

These relatons allows us to complete the rest of the relations iteratively (maybe a multiplication table would be good here)

$$\begin{aligned}
hS_1 = -iS_1^*, \quad h^2S_1 = -S_1, \quad h^3S_1 = iS_1^* \\
ghS_1 = -iS_1, \quad gh^2S_1 = -S_1^*, \quad gh^3S_1 = -iS_1 \\
hS_2 = iS_2^*, \quad h^2S_2 = -S_2, \quad h^3S_2 = iS_2^* \\
ghS_2 = iS_2, \quad gh^2S_2 = -S_2^*, \quad gh^3S_2 = -iS_2
\end{aligned}$$

Now g acts on D_{jk} in the following way

$$\begin{aligned}
g(D_{jk}f(x,y)) &= g\sum_{m,n} x^{2m+j}y^{4m+k}[x^{2m+j}y^{4m+k}]f(x,y) \\
&= \sum_{m,n}(-x)^{2m+j}y^{4m+k}f_{2m+j,4m+k} \\
&= (-1)^j D_{jk}f(x,y).
\end{aligned}$$

Similarly h acts on D_{jk} in the following way

$$h(D_jkf(x,y)) = i^k D_{jk}f(x,y).$$

Or specifically

$$gD_{00} = D_{00}, \quad gD_{11} = -D_{11}, \quad gD_{03} = D_{03}, \quad gD_{12} = -D_{12}$$
$$hD_{00} = D_{00}, \quad hD_{11} = iD_{11}, \quad hD_{03} = -iD_{03}, \quad hD_{12} = -D_{12}$$

We can now move on to finding the actions on the operator Σ. To spare some of the work it is enough to find the actions $1, gh, h^2$ and gh^3 since we'll have only 4 independent relations on the operators. Let

$$A = S_1(D_{00} + D_{12}), \quad B = S_2(D_{11} + D_{03})$$

Then

$$\Sigma = B + A, \quad gh\Sigma = B - iA$$
$$h^2\Sigma = B - A, \quad gh^3\Sigma = B + iA$$

We have

$$(1 + h^2)\Sigma = gh(1 + h^2)\Sigma = 2B$$

and

$$(1 - h^2)\Sigma = igh(1 - h^2) = 2A$$

By using these eliminations we can write new forms of the functional equation. We have

$$\frac{1}{2}(1 + h^2)(1 - t\Sigma)Q = \left(\frac{1}{2}(1 + h^2) - tB\right)Q$$

and

$$\frac{1}{2}(1 + h^2)(1) = \frac{1}{2}2 = 1$$

so the functional equation is

$$\left(\frac{1}{2}(1 + h^2) - tS_2(D_{11} + D_{03})\right)Q = 1$$

Repeating the analysis with the other relations yields

$$\left(\frac{1}{2}(1-h^2)-tS_1(D_{00}+D_{12})\right)Q=0$$

The relations allow us to write two different representations of Σ

$$2\Sigma=((1+h^2)+igh(1-h^2))\Sigma=((1-h^2)+gh(1+h^2))\Sigma$$

Or in other words

$$(1-h^2-igh(1-h^2))\Sigma=(1-h^2)(1-igh)\Sigma=0$$
$$(1+h^2-gh(1+h^2))\Sigma=(1+h^2)(1-gh)\Sigma=0$$

We can use these on the functional equation to kill Σ

$$(1-h^2)(1-igh)(1-t\Sigma)Q=(1-h^2)(1-igh)Q=0=(1-h^2)(1-igh)1$$
$$(1+h^2)(1-gh)(1-t\Sigma)Q=(1+h^2)(1-gh)Q=0=(1+h^2)(1-gh)1$$

Finally we can write the two relations as

$$(1-h^2)(1-igh)Q=(1+h^2)(1-gh)Q=0$$

It is worth mentioning that $(1-h^2)Q\neq 0$ and so on. If we use the series definition for Q here we find that these relations are saying nothing more than the fact that the terms of Q corresponding to the "holes" inside the hexagons are zero.

7.4 Finding the generating function Q

Note that

$$\frac{1}{2}(1+h^2)Q(x,y)=\frac{1}{2}(Q(x,y)+Q(x,-y))=C_1Q$$
$$\frac{1}{2}(1-h^2)Q(x,y)=\frac{1}{2}(Q(x,y)-Q(x,-y))=C_2Q$$

where the latter equalities follow since we are dealing with the even and odd y parts of Q respectively, which cover conditions C_1 and C_2. We can re-write the two functional equations from the previous section as

$$C_1Q - tS_2C_2Q = 1$$
$$C_2Q - tS_1C_1Q = 0$$

This is a linear system in the quantities C_1Q and C_2Q. The associated determinant is $1 - t^2S_1S_2 \neq 0$ so we find the unique solution to be

$$C_1Q = \frac{1}{1 - t^2S_1S_2}, \quad C_2Q = \frac{tS_1}{1 - t^2S_1S_2}$$

Finally

$$Q = C_1Q + C_2Q = \frac{1 + tS_1}{1 - t^2S_1S_2} = \sum_{k=0}^{\infty} t^{2k}S_1^kS_2^k + \sum_{k=0}^{\infty} t^{2k+1}S_1^{k+1}S_2^k$$

An expanded version of Q is

$$Q(x,y;t) = \frac{x^2y^2 + txy(x + y^2 + x^2y^2)}{x^2y^2 - t^2(1 + x^2 + xy^2)(x + y^2 + x^2y^2)}$$

An alternative solution is using the properties of $\Sigma = \Sigma_1 + \Sigma_2$ itself. First note that

$$\Sigma_1^2 = \Sigma_2^2 = 0$$

for any input because the first application of Σ_i takes a function and using C_i outputs monomials that conform to C_i. However then we multiply these by S_i which makes them no longer conform to C_i. The result follows. Therefore Σ^k consists of of a sum of k-letter words on Σ_1 and Σ_2 without consecutive appearances. There are only two such sequences - one starting with Σ_1 and the other with Σ_2. Also $\Sigma_1(1) = S_1$

but $\Sigma_2(1) = 0$. Therefore

$$\begin{aligned} Q &= \sum_{k=0}^{\infty} t^k \Sigma^k(1) = \\ &= \sum_{k=0}^{\infty} t^{2k}((\Sigma_1\Sigma_2)^k(1+t\Sigma_1) + (\Sigma_2\Sigma_1)^k(1+t\Sigma_2))(1) \\ &= \sum_{k=0}^{\infty} t^{2k}((\Sigma_1\Sigma_2)^k t\Sigma_1 + (\Sigma_2\Sigma_1)^k)(1) \end{aligned}$$

and through some more manipulation we can get

$$Q = \sum_{k=0}^{\infty} t^{2k}(1+tS_1)(\Sigma_2\Sigma_1)^k(1) = (1+tS_1)\sum_{k=0}^{\infty} t^{2k}(S_1S_2)^k$$

and the result follows.

Chapter 8

Specializing to 2 dimensions

8.1 Introduction

In a lot of ways the 2-dimensional problem looks like the 1-dimensional problem but there are major differences in the inherent complexity of the work that needs to be done. We have the functional equations

$$\begin{aligned} KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} &= \overline{a}_1\overline{a}_2 \\ K(1)Q_1^0 - K_2(1)Q_{1,2}^{0,0} &= \overline{a}_2 + a_1tR_1' \\ K(2)Q_2^0 - K_1(2)Q_{1,2}^{0,0} &= \overline{a}_1 + a_2tR_2' \\ K(1,2)Q_{1,2}^{0,0} &= 1 + a_1a_2tR_{1,2}' \end{aligned}$$

Here

$$\begin{aligned} R_1 &= S_1^{-1}Q_1^1 \\ R_2 &= S_2^{-2}Q_2^1 \\ R_{1,2} &= S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1} \end{aligned}$$

with

$$R_1' = S_1^{-1}Q_1^1 - \overline{x}_2 S_{1,2}^{-1,-1}Q_{1,2}^{1,0}$$
$$R_2' = S_2^{-1}Q_2^2 - \overline{x}_1 S_{1,2}^{-1,-1}Q_{1,2}^{0,1}$$
$$R_{1,2}' = S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}$$

We can write the functional equations in the explicit form

$$(1-tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2} \tag{8.1}$$

$$(1-ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0} - a_1t(S_1^{-1}Q_1^1 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0}) = \overline{a_2} \tag{8.2}$$

$$(1-ta_2S_2^0)Q_2^0 - (c_1 - ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0} - a_2t(S_2^{-1}Q_2^1 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}) = \overline{a_1} \tag{8.3}$$

$$(1-a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{0,-1}Q_{1,2}^{0,1}) = 1 \tag{8.4}$$

where the R_I' terms are written on the left to conserve space. We can already see that we are working in a much richer space than in the 1-dimensional case. We'll mostly work on walks with step sets that don't allow staying in one spot ($S_{1,2}^{0,0} = 0$). Sometimes we might though.

8.2 Investigating relationships between G, G_1, G_2 and $G_{1,2}$

In this work we'll focus on investigating the properties of the function $G(a_1,a_2) = Q_{1,2}^{0,0}(a_1,a_2)$. Let $G_1 = G(1,a_2)$, $G_2 = G(a_1,1)$ and $G_{1,2} = G(1,1)$. We could also see these as coefficient extractions of G with respect to b_1, b_2 and b_1b_2, i.e. $G_{1,2} = [b_1^0b_2^0]G$. In previous works it has been shown that there is a relationship between these quantities depending on the particular step set S. In some cases (like diagonal walks $S = (x_1+\overline{x_1})(x_2+\overline{x_2})$) we have a linear relationship between these quantities. In other cases (like cartesian walks $S = x_1+\overline{x}_1+x_2+\overline{x}_2$) we don't have such a linear relationship but we have a rational one, that is G is a rational function of G_1, G_2 and $G_{1,2}$. The aim of the followin sections is to investigate when such

relationships exist, when se have a relationship of that type and whether we have other cases.

The work can be split into several cases - walks that never leave the origin and ones that leave but never return to the origin. In the first case the relationship is trivially $G - G_1 - G_2 + G_{1,2} = 0$ because $Q = 1$ itself and so $G = 1$. In the second case Q may be more complicated but again $G = 1$ trivially since no walks end at the origin. This case satisfies the same relationship. However, once we move beyond this stage we need to consider the problem more carefully.

In any of the simple cases we can write $BG = 0$ and sometime $BQ = 0$ where $B = B_1B_2$ from before. We'll investigate the when we'll have a relationship like $B(FQ) = 0$, $B(FG) = 0$ or $B(FG)$ is constant for some F's.

8.3 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning

Here are all the easy cases of step sets that lead to walks that either don't leave the origin or leave it and never come back. We're not allowing resting steps here. There are 80 such sets among the 256 non-resting possible cases. Of course we could use symmetries to eliminate cases but for now this was an easier thing to do.

There are $2^5 = 32$ types of walks that don't leave the origin. Those are walks with steps among the set {NW,W,SW,S,SE}. Next we look at the walks that leave the origin but never come back. There is the case of $S = \{NE\}$ that never comes back to any axis. Let's look at those walks that never come back to the x_1 axis. We have some steps that go in the positive x_2 direction, that is one of {N, NE, E} together with some steps from {NW, W, SW}. If we were to include any other steps we would come back to the x_1 axis. If we only have a N step, we have 8 new cases, if we have only a NE step we have 3 new cases and if we have both, we have 4 new cases. This gives a total of 15 new cases. There's an equivalent number of walks that leave the x_2 axis and never come back to it. The running total is $32 + 1 + 15 + 15 = 63$.

To finish off we have 17 cases of walks that leave the origin, eventually interact with both axes but never come back to the origin. For this we must only include steps in such a way that we cannot pick a selection of steps that will decrease both

the x_1 and x_2 coordinates. We have to pick a subset of steps from {N, NE, E} so the only other steps allowed are NW and SE. The new cases coming from {N, NE, E}, {N,E}, {N, NE }, {NE, E}, {N}, {NE}, {E} are 4, 4, 2, 2, 1, 2, 2 respectively.

Before we consider walks that come back to the origin we'll look at the behavior Q depending on whether S factors in some nice way or not.

8.4 Simple case: Factoring S into a product

There's something we can say about the case where S factors as $S = S_1(x_1)S_2(x_2)$. We have

$$(DS)^k = (D_1D_2S_1S_2)^k = (D_1S_1)^k(D_2S_2)^k$$

since the D_1 and D_2 operators commute and the D_iS_i operators also commute since they deal with different variables. The functional equation for Q is

$$(1 - tDS)Q = 1$$

We can compute Q directly via the iteration as

$$Q = \sum_{k=0}^{\infty} t^k(DS)^k = \sum_{k=0}^{\infty} t^k(D_1S_1)^k(D_2S_2)^k = Q_1(x_1, a_1, t) \odot_t Q_2(x_2, a_2, t)$$

where $\odot_t$ denotes the Hadamard product of two functions with respect to variable t. From now on we'll drop the subscript t unless it is needed. Here Q_i is given by

$$Q_i = \sum_{k=0}^{\infty} t^k(D_iS_i)^k$$

and by reversing the iteration we obtain the functional equation

$$(1 - tD_iS_i)Q_i = 1$$

In other words, $Q_i(x_i, a_i, t)$ is the generating function for a 1-dimensional problem with Laurent step polynomial $S_i(x_i)$. In the case of a factoring of this sort, we are able to glue the 2-dimensional solution out of two 1-dimensional solutions. If we want to look at the generating function for quarter plane walks of length k it

is enough to look at the product of two generating functions for one-dimensional walks of length k. Also, we can write

$$Q = Q_1 \odot Q_2, \quad \widehat{Q} = \widehat{Q}_1 \odot Q_2$$
$$\widehat{Q} = Q_1 \odot \widehat{Q}_2, \quad \widehat{\widehat{Q}} = \widehat{Q}_1 \odot \widehat{Q}_2$$

Therefore

$$B_1 B_2 Q = Q - \widehat{Q} - \widehat{Q} + \widehat{\widehat{Q}} = (Q_1 - \widehat{Q}_1) \odot (Q_2 - \widehat{Q}_2)$$
$$B_1 B_2 (Q_1 \odot Q_2) = (B_1 Q_1) \odot (B_2 Q_2)$$

In general if we are applying operators that don't involve t to a Hadamard product with respect to t we can treat the operator as a constant with respect to t and apply it on whichever term we would like to. If it applies to both terms then we have to apply it to their mixture in a non-trivial way. For example, each of the Q_i's solves a 1-dimensional problem and therefore we have a relationship of the type $B_i(f_i Q_i) = g_i$ for some f_i and g_i. Then we can write

$$B_1 B_2 (f_1 Q_1 \odot f_2 Q_2) = (B_1 f_1 Q_1) \odot (B_2 f_2 Q_2) = g_1 \odot g_2$$

but so far this is not a relationship of the type $B_1 B_2 (fQ) = g$ since taking out the functions f_1 and f_2 from within the Hadamard product is non-trivial in general. One might be tempted to say that $f = f_1 \odot f_2$ but that would be incorrect. It would be neat to see if there is a way to find f in terms of f_1 and f_2.

There is a way to see the Hadamard product solution without the iteration method. Suppose that Q_i is a solution to the functional equation $(M_i - tE_iS_i)Q_i = \overline{a}_i$ for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional equation $(M - tES)Q = \overline{a}_1 \overline{a}_2$. To this end we calculate

$$\begin{aligned}(M - tES)Q &= M(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2) \\ &= (M_1 Q_1 \odot M_2 Q_2) - t((E_1 S_1 Q_1) \odot (E_2 S_2 Q_2))\end{aligned}$$

Then we substitute $M_iQ_i = \overline{a}_i + tE_iS_iQ_i$ into the first term to obtain

$$\begin{aligned}(M_1Q_1 \odot M_2Q_2) &= (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2) \\ &= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1) + (tE_1S_1Q_1) \odot (tE_2S_2Q_2))\end{aligned}$$

We have

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0$$

since we are dealing with power series in t. Therefore

$$(M_1Q_1 \odot M_2Q_2) = \overline{a}_1\overline{a}_2 + t((E_1S_1Q_1) \odot (E_2S_2Q_2))$$

and upon substitution we are left with $\overline{a}_1\overline{a}_2$ which is what we needed in the first place.

One may also ask whether we can do something in the case where S factors and the terms share variables.

This factoring also generalizes to higher dimensions. If S factors into a product of k factors $S_1, \ldots, S_k$ where no two of which share any variables then the problem of finding Q boils down to solving k lower-dimensional problems for functions $Q_1, \ldots, Q_k$, naming variables appropriately and taking the Hadamard product of the functions. That is

$$Q = Q_1 \odot \cdots \odot Q_k$$

The more lower-dimensional problems that we solve, the more higher dimensional problems we can solve. Although it appears that the interesting higher dimensional problems that can be solved in this manner will have a lower and lower proportion among all higher dimensional problems.

8.5 Kernel Group Considerations

It is important to find more functional equations if we can. One way to do this is by applying a transformation on the pair (x_1, x_2) that leaves as many parts of the functional equation as we can but still adding some information in the form of a new equation. One way to do this is to cosider mapping x_1 and x_2 to some quantities separately and then consider the full group generated by such maps. Since we

usually eliminate the kernel in front of Q it is useful to consider transformations that preserve the kernel. We'll call this group $G(K)$, although it won't be the full group of rational transformations preserving the kernel. For example $(x_1,x_2) \mapsto (x_2,x_1)$ will not be in the group. Any transformation on (x_1,x_2) that preserves the kernel $K = 1 - tS$ must necessarily preserve the step polynomial S. Here

$$S = \alpha\overline{x} + \beta + \gamma x$$

where $x = x_1$ or x_2 and α, β, γ change correspondingly. We consider sending $x \mapsto X$ and solve for X to determine the possible transformations we can perform. That is we have

$$\begin{aligned}
\alpha\overline{x} + \gamma x &= \alpha\overline{X} + \gamma X \\
\alpha(\overline{x} - \overline{X}) &= \gamma(X - x) \\
\alpha\frac{X - x}{Xx} &= \gamma(X - x)
\end{aligned}$$

We have cases

- $\alpha = 0, \gamma = 0$. The equation is then trivially true and any transformation preserves the kernel although this is not very useful though.
- $\alpha = 0, \gamma \neq 0$. Then $X = x$ and only the identity transformation preserves the kernel.
- $\alpha \neq 0, \gamma = 0$. Here $X = x$ as well.
- $\alpha \neq 0, \gamma \neq 0$. Here we either have $X = x$ or $X \neq x$ and $X = \overline{x}\alpha/\gamma = \overline{x}S_{-1}/S_1$

In conclusion, either every transformation works and we don't have an interesting case or we have the identity transformation plus a possible second one. The last transformation creates a non-trivial group and we can have groups of various orders. Let

$$g_1 : x_1 \mapsto \overline{x_1}\frac{S_1^{-1}}{S_1^1}, \quad g_2 : x_2 \mapsto \overline{x_2}\frac{S_2^{-1}}{S_2^1}$$

whenever the S quantities are non-zero. Note that $g_i(g_i(x_1)) = x_1$, that is they are involutions. We have three cases for each x

- 1: Any transformation can be applied
- 2: We only have the identity transformation
- 3: We have two transformations

We'll combine cases for analysis in the form "case $i.j$" to mean we have case i for x_1 and case j for x_2.

Case 1.1: Here we can apply any transformation on both x_1 and x_2. From the transformation analysis this means that $S_1^{-1} = S_1^1 = S_2^{-1} = S_2^1 = 0$ and thus means that we have no steps that go away from the origin. Therefore $S = S_{1,2}^{0,0}$. Thus we have only two models - no steps or a resting step. (I should include little diagrams with arrows like before).

Case 1.2 : Here any transformation applies on x_1, so $S_1^{-1} = S_1^1 = 0$ so the model has no steps that step away from the origin in the x_1-direction. Since the only transformation on x_2 is the identity transformation we either have $S_2^{-1} = 0$ or $S_2^1 = 0$ in which case the model consists of only an up step or a down step, together with a possible resting step. Similarly, the 2.1 case involves a single x_1 or $\overline{x_1}$ step with a possible resting step.

Case 1.3: (and 3.1): This case is similar to the previous one except that now we have two transformations for x_2 and this means $S_2^{-1}, S_2^1 \neq 0$. This means we must have either both vertical steps x_2 and $\bar{x}_2$ in the 1.3 case or two horizontal steps x_1 and $\bar{x}_1$ in the 3.1 case.

Case 2.2: In this case both transformations are the identity transformations. Therefore exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. WLOG, assume that $S_1^1 \neq 0, S_2^1 \neq 0$ (the other models can be obtained by a rotation around the origing of 90, 180 and 270 degrees). The possible models are (I should include the drawings): {N,E,NE}, {N,NE}, {E,NE}, {N,E}, {NE}. In other words, the subsets of {N,E,NE} except the ones going either only in the N direction or only in the E direction. This is a case of group $G(K)$ or order 1.

Case 2.3: In this case there is only one transformation for x_1 and two for x_2. This means $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case). Also $S_2^{-1}, S_2^1 \neq 0$ so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. We have 17 cases (and another 17 including a resting step). Of course in the 3.2 case we have the meaning of x_1 and x_2 reversed.This is a case of group $G(K)$ of order 2.

Case 3.3: This is the most interesting case with two transformations for x_1 and x_2 each. This means that there must be at least one step going in a positive and one going in a negative coordinate for each of x_1 and x_2. There are a lot of such cases (in the hundreds). Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity depending on the step set in question.

8.6 Solving basic 2D models

8.6.1 Cases 1.1, 1.2, 1.3

A quick glance at the step models involved shows that we are dealing with essentially one-dimensional problems. This means they can be solved easily explicitly. One can also note that these models are all factorable

$$S(x_1, x_2) = 1 \cdot S(x_2)$$

Both Q_1 and Q_2 are functions we solved for in the one-dimensional case

$$Q_1(x_1, a_1, t) = \frac{1}{1 - a_1 t}, \quad Q_2(x_2, a_2, t) = q(x_2, a_2, t)$$

for some solution q from that section. Therefore

$$Q(x_1, x_2, a_1, a_2, t) = Q_1(x_1, a_1, t) \odot Q_2(x_2, a_2, t) = \frac{1}{1 - a_1 t} \odot q(x_2, a_2, t) = q(x_2, a_2, a_1 t)$$

Similarly in the 2.1, 3.1 cases

$$Q(x_1, x_2, a_1, a_2, t) = Q_1(x_1, a_1, t) \odot Q_2(x_2, a_2, t) = q(x_1, a_1, t) \odot \frac{1}{1 - a_2 t} = q(x_1, a_1, a_2 t)$$

and this completes the solution. What about the relationship with the B_1B_2 transformation? If $f(a,t)$ is the function that goes together with $q(x,a,t)$ then the function $F(a_1,a_2,t)$ that goes together with $Q(x_1,x_2,a_1,a_2,t)$ is given by

$$F(a_1,a_2,t) = f(a_2,a_1t)$$

Then we have $B_2(FQ) = 0$ and $B_1B_2(FQ) = 0$.

8.6.2 Case 2.2 - group of order 1

A quick glance at the step models reveals that this case is mostly trivial. We have a lot of models that never leave the origin and those have a trivial solution

$$Q(x_1,x_2,a_1,a_2) = \frac{1}{1-a_1a_2S^{0,0}_{1,2}t}.$$

We also have a lot of models that are essentially one-dimensional with a single step going along an axis and some steps going off-axis where the latter steps can never be taken by a random walk in this model. They are essentially cases already covered in the previous section. Suppose that the single step is taken in the positive x_1 direction. This is equivalent to the $s_{-1} = 0$ case in 1D with function

$$q(x,a,t) = \frac{1-s_0t}{(1-st)(1-as_0t)} = \frac{1-s_0t}{(1-x_1s_1t)(1-as_0t)}$$

Then we have

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = q(x_1,a_1,t) \odot \frac{1}{1-a_2t} = q(x_1,a_1,a_2t)$$

and therefore

$$Q(x_1,x_2,a_1,a_2,t) = \frac{1-a_2S^{0,0}_{1,2}t}{(1-(x_1+S^{0,0}_{1,2})a_2t)(1-a_1a_2S^{0,0}_{1,2}t)}$$

The interesting models in case 2.2 are those with non-negative steps that we listed. Here

$$S = \alpha x_1 + \beta x_2 + \gamma x_1x_2 + \delta$$

The functional equation and its zero extractions are

$$\begin{aligned}(1-tS)Q - c_1Q_1^0 - c_2Q_2^0 + c_1c_2Q_{1,2}^{0,0} &= \overline{a_1a_2}\\ (1-ta_1(\delta+\beta x_2))Q_1^0 - c_2Q_{1,2}^{0,0} &= \overline{a_2}\\ (1-ta_2(\delta+\alpha x_1))Q_2^0 - c_1Q_{1,2}^{0,0} &= \overline{a_1}\\ (1-a_1a_2t\delta)Q_{1,2}^{0,0} &= 1\end{aligned}$$

We solve the problem by a simple back-substitution of the variables.

$$\begin{aligned}Q_{1,2}^{0,0} &= \frac{1}{1-a_1a_2\delta t}\\ Q_1^0 &= \frac{\overline{a}_2}{1-ta_2S_1^0}\left(1+\frac{b_2}{1-a_1a_2\delta t}\right) = \frac{1-a_1\delta t}{(1-ta_1S_1^0)(1-a_1a_2\delta t)}\\ Q_2^0 &= \frac{\overline{a}_1}{1-ta_1S_2^0}\left(1+\frac{b_1}{1-a_1a_2\delta t}\right) = \frac{1-a_2\delta t}{(1-ta_2S_2^0)(1-a_1a_2\delta t)}\\ Q &= \frac{1}{1-tS}(\overline{a}_1\overline{a}_2 + c_1Q_1^0 + c_2Q_2^0 - c_1c_2Q_{1,2}^{0,0})\\ &= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-ta_1a_2\delta}\left(c_1\frac{1-a_1\delta t}{1-ta_1S_1^0} + c_2\frac{1-a_2\delta t}{1-ta_2S_1^0} - c_1c_2\right)\right)\\ &= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-a_1a_2\delta t}(c_1f_1(a_1,t) + c_2f_2(a_2,t) - c_1c_2)\right)\end{aligned}$$

where the last line explicitly shows that Q is composed of functions that depend only on one of a_1, a_2. If we want to find a relation of the type $B(FQ) = 0$ then we must first find a suitable function F. First of all the $1-tS$ part doesn't play a role since it doesn't depend on a_1 or a_2. Then later the variables c_1 and c_2 will vanish when extracting the coefficients of b_1^0 and b_2^0. We'll now show that the function

$$F(a_1,a_2,t) = 1-a_1a_2\delta t$$

works exactly. First

$$B((1-a_1a_2\delta t)\overline{a}_1\overline{a_2}) = B(\overline{a}_1\overline{a}_2 - t\delta) = B(\overline{a}_1\overline{a}_2) - B(t\delta) = \overline{a}_1\overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1c_2$$

Then

$$B(c_1f_1 + c_2f_2 - c_1c_2) = c_1f_1 + c_2f_2 - c_1c_2 - c_2f_2 - c_1f_1 = -c_2c_2$$

Putting the two quantities together along with dividing by $1-tS$ shows that $B(FQ) = 0$. Of course this works for $G = Q^{0,0}_{1,2}$ too.

8.6.3 Case 2.3 - group of order 2

There are two possibilities here - positive and negative. In the negative case $S^1_1 = 0$ and the steps have a non-positive x_1 component. Due to the fact that a lot of steps are not going to be taken by random walks in those models, all of these models reduce to the cases of S consising of a rest step and maybe a single step in a positive direction. This takes care of 34 models (or 68 if we include a rest step). The interesting models are those with a positive x_1 component, or in other words, models where $S^{-1}_1 = 0$. The calculations for these models become increasingly more complicated. The functional equation and its extracted forms read as

$$\begin{aligned}
(1-tS)Q - c_1Q^0_1 - (c_2 - t\bar{x}_2S^{-1}_2)Q^0_2 + c_1c_2Q^{0,0}_{1,2} &= \overline{a_1a_2} \\
(1-ta_1S^0_1)Q^0_1 - (c_2 - ta_1\bar{x}_2S^{0,-1}_{1,2})Q^{0,0}_{1,2} &= \overline{a_2} \\
(1-ta_2S^0_2)Q^0_2 - c_1Q^{0,0}_{1,2} - a_2tS^{-1}_2Q^1_2 &= \overline{a_1} \\
(1-a_1a_2tS^{0,0}_{1,2})Q^{0,0}_{1,2} - ta_1a_2S^{0,-1}_{1,2}Q^{0,1}_{1,2} &= 1
\end{aligned}$$

This is starting to look somewhat daunting. However, we can identify a subproblem that is essentially one-dimensional and therefore already solved. For this we look at the second and fourth equations with the substitution $a_i = a, c_i = c, x_2 = x, ta_1 = \tau, S^0_1 = T, Q^0_1(x_2) = q(x)$. These equations become

$$\begin{aligned}
(1-\tau T)q - (c - \tau\bar{x}T_{-1})q_0 &= \overline{a} \\
(1-a\tau T_0)q_0 - a\tau S^{0,-1}_{1,2}q_1 &= 1
\end{aligned}$$

where $q_0 = q(a,\tau)$. This is the one-dimensional problem again. The solution here is $q(x,a,\tau)$ so the original problem has a solution

$$Q_1^0(x_2) = q(x_2, a_2, a_1 t), \quad Q_{1,2}^{0,0} = q_0(a_2, a_1 t).$$

The reason the method worked in this case is that we have steps that keep a random walk staying on the x_2 axis and steps that take it away from the x_2 but no steps that have it return to it. This is similar to the case where a random walk leaves the x_2 axis never to come back. Essentially the problem splits into a one-dimensional one regarding the x_2 axis and a further 2D one once we step off the axis. We re-write the functional equation as

$$(1-tS)Q - (c_2 - t\bar{x}_2 S_2^{-1})Q_2^0 = \overline{a}_1 \phi(x_2) = \overline{a_1}(\overline{a}_2 + b_1 Q_1^0 - b_1 c_2 Q_{1,2}^{0,0})$$

Here we can treat x_2 as a constant and solve for a one-dimensional problem with x_1 as a variable. We can perform the same analysis as in the the one-dimensional case except where we had $\overline{a}_1$ we would now have $\overline{a}_1\phi(x_2)$. Therefore we can solve for Q explicitly. However I haven't done this yet so I'll leave this for later. Curiously enough this analysis does not use the fact that we are dealing with a group of order 2 anymore than the one-dimensional case uses the fact that $|G(K)| = 2$ for $K = 1 - t(\bar{x} + x)$. Interestingly we can try re-doing the problem using the group structure. It would surely illustrate the methods we would use later in analysing the higher group size models. I've done it on paper and I don't think I've made any mistakes. However I don't obtain the correct answers. I suspect it's because of the following - $\sigma_1 = 1/(tS_1^1) - S_1^0/S_1^1$ which does not expand as a power series in t. Perhaps we just cannot plug this solution into the functional equations. Therefore one would have to resort to using the σ_2 solution without being able to use the group structure at all. I can certainly solve the problem in the above manner and use the series generated for each part of the solution to test whether the second solution really fails, and if it does test if it fails because of the σ_1 behavior.

Chapter 9

Highly-symmetric 2D models

When we consider the supersymmetric cases the x-transformations are given by

$$\bullet : x_1 \mapsto \overline{x_1}, \quad * : x_2 \mapsto \overline{x_2}$$

In each case we have

$$K^\bullet = K^* = K, \quad K_2^\bullet = K_2, \quad K_1^* = K_1$$

while the others change in some way. The solutions eliminating the kernel also have a nice form

$$\sigma_i = \frac{1 - S_i^0 - \sqrt{(1-S_i^0)^2 - 4t^2(S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1 - S_i^0 + \sqrt{(1-S_i^0)^2 - 4t^2(S_i^1)^2}}{2tS_i^1}$$

because we have

$$\overline{\sigma}_i = \frac{S_i^1}{S_i^{-1}} conj(\sigma_i) = conj(\sigma_i)$$

We can take the master equation and apply $*$ to it

$$-KQ + K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0$$
$$-KQ^* + K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0$$

From here on we will input $x_1 = \sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0$$
$$K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1 K_2^* Q_1^0 + K_2^* K_2 Q_2^0 - K_2^* K_{1,2} Q_{1,2}^{0,0} + \alpha K_2^* = 0$$
$$K_1 K_2 Q_1^{0,*} + K_2 K_2^* Q_2^0 - K_2 K_{1,2}^* Q_{1,2}^{0,0} + \alpha K_2 = 0$$

We subtract these two equations to obtain

$$K_1(K_2^* Q_1 - K_2 Q_1^*) - (K_2^* K_{1,2} - K_2 K_{1,2}^*) Q_{1,2} + \alpha(K_2^* - K_2) = 0$$
$$K_2^* Q_1 - K_2 Q_1^* - \frac{K_2^* K_{1,2} - K_2 K_{1,2}^*}{K_1} Q_{1,2} + \alpha \frac{K_2^* - K_2}{K_1} = 0$$

The differences in solutions will occur from here on out. We have

$$K_2^* - K_2 = c_2 - t x_2 S_2^{-1} - c_2 + t\overline{x_2} S_2^{-1} = t(\overline{x_2} - x_2) S_2^{-1}(\sigma_1)$$

Next

$$\begin{aligned} K_2^* K_{1,2} - K_2 K_{1,2}^* &= (c_2 - t x_2 S_2^{-1})(c_1 c_2 - t\overline{x_1 x_2} S_{1,2}^{-1,-1}) - (c_2 - t\overline{x_2} S_2^{-1})(c_1 c_2 - t\overline{x_1} x_2 S_{1,2}^{-1,-1}) \\ &= c_1 c_2 t S_2^{-1}(\overline{x_2} - x_2) - c_2 t \overline{x_2} S_{1,2}^{-1,-1}(\overline{x_2} - x_2) \\ &= c_2 t(\overline{x_2} - x_2)(c_1 S_2^{-1}(\sigma_1) - \overline{x_1} S_{1,2}^{-1,-1}) \end{aligned}$$

9.1 Cartesian model: steps N, S, E, W

In this case $K_{1,2} = c_1 c_2$ so we have a simpler situation where we can re-write the above equation as

$$K_2^* Q_1 - K_2 Q_1^* + \alpha\beta \frac{K_2^* - K_2}{K_1} = 0$$

where

$$\beta = 1 - b_1 b_2 Q_{1,2}^{0,0}$$

Furthermore

$$K_2^* - K_2 = t(\overline{x_2} - x_2)$$

So we can let

$$f = f(c_1, x_2) = \frac{t(\overline{x_2} - x_2}{K_1}$$

since K_1 doesn't depend on c_2 or x_1. The equation finally becomes

$$(c_2 - t x_2) Q_1^0(x_2) - (c_2 - t\overline{x_2}) Q_1^0(\overline{x_2}) + \alpha\beta f = 0$$

We perform the extraction $[x_2^1]$ to get

$$c_2 Q_{1,2}^{0,1} - t Q_{1,2}^{0,0} + \alpha\beta f_2^1 = 0$$

and the extraction $[b_2^0]$ to get

$$-t\widehat{Q}_{1,2}^{0,0} + \overline{a_1} f_2^1$$

Using the second equation within the first one and simplifying yields

$$b_2 Q_{1,2}^{0,1} = t(a_2 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0})$$
$$b_1 Q_{1,2}^{1,0} = t(a_1 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0})$$

Here the second equation is obtained via the same method but with $\bullet$ instead of $*$. Using the same method we can also obtain

$$b_2 Q_{1,2}^{1,1} = t(a_2 Q_{1,2}^{1,0} - \beta \widehat{Q}_{1,2}^{1,0})$$

Multiplying by b_1 and using the above relations leads to

$$\begin{aligned} b_1b_2Q^{1,1}_{1,2} &= t(a_2b_1Q^{1,0}_{1,2} - \beta b_1\widehat{Q}^{1,0}_{1,2}) \\ &= ta_2t(a_1Q^{0,0}_{1,2} - \beta\widehat{Q}^{0,0}_{1,2}) - t\beta t(a_1\widehat{Q}^{0,0}_{1,2} - \beta\widehat{\widehat{Q}}^{0,0}_{1,2}) \\ &= t^2(a_1a_2Q^{0,0}_{1,2} - a_2\beta\widehat{Q}^{0,0}_{1,2} - a_1\beta\widehat{Q}^{0,0}_{1,2} + \beta^2\widehat{\widehat{Q}}^{0,0}_{1,2}) \end{aligned}$$

Let

$$G = Q^{0,0}_{1,2}, \quad \widehat{G} = G_1, \quad \widehat{G} = G_2$$

From before we have

$$Q^{0,0}_{1,2} = 1 + ta_1a_2(Q^{1,0}_{1,2} + Q^{0,1}_{1,2})$$

Multiplying by b_1b_2 and using the above relations leads to

$$\begin{aligned} b_1b_2G &= b_1b_2 + ta_1a_2(b_2b_1Q^{1,0}_{1,2} + b_1b_2Q^{0,1}_{1,2}) \\ &= b_1b_2 + ta_1a_2(b_2t(a_1G - \beta G_1) + b_1t(a_2G - \beta G_2) \\ &= b_1b_2 + t^2a_1a_2(b_2a_1G + b_1a_2G - b_2\beta G_1 - b_1\beta G_2) \end{aligned}$$

Using the relation for β and simplifying leads

$$G = \frac{b_1b - 2 - t^2a_1a_2(b_2G_1 + b_1G_2)}{b_1b_2 - t^2a_1a_2(a_2b_1 + a_1b_2 + b_1b_2(b_2G_1 + b_1G_2))}$$

If we let $F = (b_2G_1 + b_1G_2)$ and perform the long division with respect to F we find that

$$Q^{0,0}_{1,2} = \frac{1}{b_1b_2}\left(1 - \frac{b_1b_2(1 - b_1b_2) - t^2a_1a_2(a_1b_2 + a_2b_1)}{b_1b_2 + t^2a_1a_2(a_1b_2 + a_2b_1 + b_1b_2F)}\right)$$

Using arguments similar to those in the later section on diagonal steps we can find out more about the various Q's.

$$\begin{aligned} b_2Q^{m,n+1}_{1,2} &= t(a_2Q^{m,n}_{1,2} - \beta\widehat{Q}^{m,n}_{1,2}) \\ b_1Q^{m+1,n}_{1,2} &= t(a_1Q^{m,n}_{1,2} - \beta\widehat{Q}^{m,n}_{1,2}) \end{aligned}$$

Summing the equations multiplied by their respective powers of x_1 and x_2 we find the equations

$$b_2 Q_2^{n+1} = t(a_2 Q_2^n - \beta \widehat{Q}_2^n)$$
$$b_1 Q_1^{m+1} = t(a_1 Q_1^m - \beta \widehat{Q}_1^m)$$

Now we can sum with respect to x_2 and x_1 to find

$$b_2(Q - Q_2^0) = x_2 t(a_2 Q - \beta \widehat{Q})$$
$$b_1(Q - Q_1^0) = x_1 t(a_1 Q - \beta \widehat{Q})$$

We can solve these for Q_2^0 and Q_1^0 as follows

$$b_2 Q_2^0 = (b_2 - a_2 x_2 t)Q + x_2 t \beta \widehat{Q}$$
$$b_1 Q_1^0 = (b_1 - a_1 x_1 t)Q + x_1 t \beta \widehat{Q}$$

We can substitute these two expressions into the original functional equation and after some algebra we find that

$$Q = \frac{(1 - b_1 b_2 G)(\alpha + t x_1 \overline{b_1} K_1 \widehat{Q} + t x_2 \overline{b_2} K_2 \widehat{Q})}{K - (1 - \overline{c_1} x_1 t)K_1 - (1 - \overline{c_2} x_2 t)K_2}$$

and after recognizing some of the quantities as modified K_1 and K_2's we finally stop at the solution for Q

$$Q = \frac{(1 - b_1 b_2 G)(\alpha - \overline{b}_1 K_1 \widehat{K}_1^{\bullet} \widehat{Q} - \overline{b}_2 K_2 \widehat{K}_2^{*} \widehat{Q})}{K - \overline{c}_1 K_1 K_1^{\bullet} - \overline{c}_2 K_2 K_2^{*}}$$

This expression is as perplexing as it is complicated. Why should the K's be combined in such a ridiculous way in order to give Q in terms of

9.2 Diagonal model: steps NE, NW, SE, SW

In this case the roots of the kernel become

$$\sigma_i = \frac{1-\sqrt{1-4t^2(S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1+\sqrt{1-4t^2(S_i^1)^2}}{2tS_i^1}$$

with the extra property that

$$\sigma_1 + \overline{\sigma_1} = \frac{1}{tS_1^{-1}}.$$

First we have

$$\begin{aligned} K_2^* - K_2 &= t(\overline{x_2} - x_2)S_2^{-1}(\sigma_1) \\ &= t(\overline{x_2} - x_2)(\overline{\sigma_1} + \sigma_1) \\ &= \frac{\overline{x_2} - x_2}{\overline{x_2} + x_2} \end{aligned}$$

We have

$$\begin{aligned} K_2^* K_{1,2} - K_2 K_{1,2}^* &= (c_2 - tx_2S_2^{-1})(c_1c_2 - t\overline{\sigma_1}\overline{x_2}) - (c_2 - t\overline{x_2}S_2^{-1})(c_1c_2 - t\overline{\sigma_1}x_2) \\ &= c_1c_2t(\overline{x_2} - x_2)S_2^{-1} - c_2t\overline{\sigma_1}(\overline{x_2} - x_2) \\ &= c_2t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \end{aligned}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1-\sqrt{1-4t^2v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1+\sqrt{1-4t^2v^2}}{2tv}$$

We have

$$\begin{aligned} (c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) &= \frac{c1}{tv} - \frac{1+\sqrt{1-4t^2v^2}}{2tv} \\ &= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1-\sqrt{1-4t^2v^2}}{2}\right) \end{aligned}$$

Also

$$\begin{aligned}
K_1 &= c_1 - tv\overline{\sigma_1} \\
&= c_1 - tv\frac{1+\sqrt{1-4t^2v^2}}{2tv} \\
&= 1 - \overline{a_1} - \frac{1+\sqrt{1-4t^2v^2}}{2} \\
&= -\overline{a_1} + \frac{1-\sqrt{1-4t^2v^2}}{2}
\end{aligned}$$

Thus we finally have

$$\begin{aligned}
\frac{1}{K_1}(K_2^*K_{1,2} - K_2K_{1,2}^*) &= -\frac{c_2t(\overline{x_2}-x_2)(c_1(\sigma_1+\overline{\sigma_1})-\overline{\sigma_1})}{c_1 - t\overline{\sigma_1}v} \\
&= -\frac{c_2t(\overline{x_2}-x_2)(-1)}{tv} \\
&= \frac{c_2(\overline{x_2}-x_2)}{\overline{x_2}+x_2}
\end{aligned}$$

Thus the equation will now read as

$$K_2^*Q_1 - K_2Q_1^* - \frac{c_2(\overline{x_2}-x_2)}{\overline{x_2}+x_2}Q_{1,2} + \alpha\frac{\overline{x_2}-x_2}{(\overline{x_2}+x_2)K_1} = 0$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2}+x_2$. We have

$$\begin{aligned}
R^* &= (\overline{x_2}+x_2)K_2^* = c_2(\overline{x_2}+x_2) - x_2 = c_2\overline{x_2} - \overline{a_2}x_2 \\
R &= (\overline{x_2}+x_2)K_2 = c_2(\overline{x_2}+x_2) - \overline{x_2} = c_2x_2 - \overline{a_2x_2}
\end{aligned}$$

We have a couple of relations here

$$\begin{aligned}
R^* - R &= v(K_2^* - K_2) = \overline{x_2} - x_2 \\
\overline{x_2}R^* - x_2R &= \overline{x_2}c_2v - 1 - x_2c_2v + 1 \\
&= c_2v(\overline{x_2} - x_2)
\end{aligned}$$

The equation becomes

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^0(x_2) + (\overline{a_2x_2} - c_2x_2)Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + (\overline{x_2} - x_2)\overline{K}_1\alpha = 0$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 = 0$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x_1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1$$

We can extract higher coefficients too. Let $k \geq 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0}) \end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction.

We have

$$Q_1^0 = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1$$
$$Q_1^{0,*} = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}$$

Combining them with equation (9.2) gives

$$\begin{aligned}
0 &= R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\
&= R^*(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1) - R(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2 Q_{1,2}^{0,0}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0)
\end{aligned}$$

So we can write the equation

$$R^* Q_1^1 - R Q_1^{1,*} + \overline{a_2}\delta_1 = 0$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1 tv}$$

This equation together with equation (9.2) form the base cases for the following lemma

Lemma 2. Let $m \geq 0$ be an integer. Then

$$R^* Q_1^m - R Q_1^{m,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{9.1}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1,0)$. Now we assume that the formula is true for all cases up to

and including m. Then we use formula (9.2)

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*} \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*}) \\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x_2}R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2)Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1}) \\
&+ c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0}) - \overline{a_2}tv\delta_{m-1})
\end{aligned}$$

Combining this with equation (9.1) leads to

$$\begin{aligned}
0 &= tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} - \overline{a_2}tv\delta_{m-1} + \overline{a_2}\delta_m \\
0 &= R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} + \overline{a_2}\delta_{m+1}
\end{aligned}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1}$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a_2x_2})Q_1^m(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0$$

We start by extracting the $[x_2^1]$ coefficient

$$\begin{aligned}
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} - c_2Q_{1,2}^{m,0} + c_2Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0 \\
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0
\end{aligned}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geq 2$

$$c_2Q_{1,2}^{m,n+2} - \overline{a_2}Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a_2}\delta_m = 0$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\widehat{Q}^{m,n}_{1,2} + [x_2^{n+1}]\delta_m = 0$$

Substituting this back into the above equation gives

$$b_2 Q^{m,n+2}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 3. For integers $m, n \geq 0$ we have the following relations

$$b_1 Q^{m+2,n}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}$$
$$b_2 Q^{m,n+2}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}$$

We can sum up those equations to obtain functional equations

Corollary 4. For integers $m \geq 0$ we have the following relations

$$Q^m_1 - \widehat{Q}^m_1 = b_1 Q^{m+2}_1$$
$$\widehat{Q}^m_1 - \widehat{\widehat{Q}}^m_1 = b_1 \widehat{Q}^{m+2}_1$$
$$x_1^2(Q^n_2 - \widehat{Q}^n_2) = b_1(Q^n_2 - Q^{1,n}_{1,2}x_1 - Q^{0,n}_{1,2})$$
$$x_1^2(\widehat{Q}^n_2 - \widehat{\widehat{Q}}^n_2) = b_1(\widehat{Q}^n_2 - \widehat{Q}^{1,n}_{1,2}x_1 - \widehat{Q}^{0,n}_{1,2})$$
$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q^1_1 - Q^0_1)$$
$$x_1^2(\widehat{Q} - \widehat{\widehat{Q}}) = b_1(\widehat{Q} - x_1 \widehat{Q}^1_1 - \widehat{Q}^0_1)$$

Corollary 5. For integers $n \geq 0$ we have the following relations

$$\begin{aligned}
Q_2^n - \widehat{Q}_2^n &= b_2 Q_2^{n+2} \\
\widehat{Q}_2^n - \widehat{\widehat{Q}}_2^n &= b_2 \widehat{Q}_2^{n+2} \\
x_2^2(Q_1^m - \widehat{Q}_1^m) &= b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \\
x_2^2(\widehat{Q}_1^m - \widehat{\widehat{Q}}_1^m) &= b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \\
x_2^2(Q - \widehat{Q}) &= b_2(Q - x_2 Q_2^1 - Q_2^0) \\
x_2^2(\widehat{Q} - \widehat{\widehat{Q}}) &= b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0)
\end{aligned}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q_{1,2}^{0,0}$

Now we can move onto the main result about the diagonal walks and it is that the function for labelled walks ending at $(0,0)$ is connected to the functions for semi-labelled and unlabelled walks endind at $(0,0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients yielding the following two relations

$$\begin{aligned}
Q_{1,2}^{0,0} &= 1 + t a_1 a_2 Q_{1,2}^{1,1} \\
Q_{1,2}^{1,1} &= t(Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2})
\end{aligned}$$

We also have

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,2} &= b_1(Q_{1,2}^{2,0} - \widehat{Q}_{1,2}^{2,0}) \\
&= b_1 Q_{1,2}^{2,0} - b_1 \widehat{Q}_{1,2}^{2,0} \\
&= Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} + \widehat{\widehat{Q}}_{1,2}^{0,0}
\end{aligned}$$

together with

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,0} &= b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} \\
b_1 b_2 Q_{1,2}^{0,2} &= b_1 Q_{1,2}^{0,0} - b_1 \widehat{Q}_{1,2}^{0,0}
\end{aligned}$$

We we have

$$\begin{aligned}
b_1b_2Q^{1,1}_{1,2} &= t(b_1b_2Q^{0,0}_{1,2} + b_1b_2Q^{2,0}_{1,2} + b_1b_2Q^{0,2}_{1,2} + b_1b_2Q^{2,2}_{1,2}) \\
&= t(b_1b_2Q^{0,0}_{1,2} + b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} + b_1Q^{0,0}_{1,2} - b_1\widehat{Q}^{0,0}_{1,2} + Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \\
&= t((b_1b_2 + b_1 + b_2 + 1)Q^{0,0}_{1,2} - (b_2+1)\widehat{Q}^{0,0}_{1,2} - (b_1+1)\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \\
&= t(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2})
\end{aligned}$$

We can finally write

$$\begin{aligned}
b_1b_2Q^{0,0}_{1,2} &= b_1b_2 + ta_1a_2b_1b_2Q^{1,1}_{1,2} \\
&= b_1b_2 + t^2a_1a_2(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2})
\end{aligned}$$

Let $G(b_1,b_2)$ stand for $Q(0,0,b_1,b_2)$ and G_i has $b_i = 0$. Then we can write it all as

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2})$$

or written as a linear combination

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2$$

We can divide out by a_1a_2 and obtain the equation

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2$$

from which we can read off

$$f(a_1,a_2,t) = c_1c_2 - a_1a_2t^2$$

and

$$G = \frac{(c_1c_2 - t^2(a_2G_1 + a_1G_2 - G_{1,2})}{f(a_1,a_2,t)}$$

together with $B(fG) = c_1c_2$ Also, we can solve for Q^j_i and friends, in terms of various $[b^l_k]$ extractions, so overall it should lead to a relationship of the form

$B(FQ) = 0$.

9.3 The almost-full model: missing steps E and W

9.4 The full model: all steps available

9.5 Conjectured relationships depending on $S_{1,2}^{-1,-1}$

As we have seen whenever $S_{1,2}^{-1,-1} = 0$, α and G tend to go together in the equations, so in the end we have a solution that depends on $\beta(G)$ multiplied by G_1, G_2 and/or $G_{1,2}$. In the end this leads to a rational function solution for G in terms of G_1, G_2 and $G_{1,2}$. It's then easy to conjecture that this happens whenever we can have such an algebraic relationship. The other part of the conjecture is that we'd have a linear relationship between G, G_1, G_2 and $G_{1,2}$ when $S_{1,2}^{-1,-1} \neq 0$.

Chapter 10

Higher Group Orders - 4, 6 and 8

10.1 Nonsymmetric group of order 4

10.2 Group of order 6

10.3 Group of order 8

Chapter 11

Epilogue

11.1 Discussion

In this work we examined such and such things. This led to so and so stuff. Finally we see problems in this and that area. Some things remain a mystery.

11.2 Future Work

Obviously we need to solve the remaining problems. We also need to solve/explain the mysteries presented so far. Additionally we need to rigourize the subject some more and bring in some more abstraction to make the methods more powerful. Full treatment of flarvoured problems.

11.3 Conclusion

This work solved the challenges it presented itself and uncovered more challenges that await a good tackling.

Chapter 12

Miscellaneous Things

If we could prove the validity of one of the functional equations in the corollaries (5th row in each corollary) we would then be able to extract coefficients and obtain the formulas in the first corollary. However, since we have a direct proof already, this section is kind of irrelevant now. We'll just keep it for now because it has some interesting math. The steps from now on are almost exactly the same as those that you'd have to do to prove the Dyck path formula. We'll prove the last one involving a b_1 coefficient. First we have

$$\begin{aligned} K &= 1 - tS = 1 - t(x_1 + \overline{x_1})(x_2 + \overline{x_2}) \\ K_1 &= c_1 - t\overline{x_1}S_1^{-1} = c_1 + \widehat{K}_1 \\ K_2 &= c_2 - t\overline{x_2}S_2^{-1} \\ K_{1,2} &= c_1c_2 - t\overline{x_1x_2} = c_1c_2 + \widehat{K}_{1,2} \end{aligned}$$

We also need the coefficient extractions

$$\begin{aligned} [x_1^0]: \quad & Q_1^0 - c_2Q_{1,2}^{0,0} - ta_1S_1^{-1}Q_1^1 = \overline{a_2} \\ [x_1^0]: & Q_1^1 - tS_1^{-1}Q_1^2 - tS_1^{-1}Q_1^0 + t\overline{x_2}Q_{1,2}^{2,0} + t\overline{x_2}Q_{1,2}^{0,0} = 0 \\ & -t\overline{x_1}S_1^{-1}Q_1^2 + t\overline{x_1x_2}Q_{1,2}^{2,0} = t\overline{x_1}S_1^{-1}Q_1^0 - t\overline{x_1x_2}Q_{1,2}^{0,0} - \overline{x_1}Q_1^1 \end{aligned}$$

which we'll use soon. We first start with

$$\begin{aligned}
KQ &= K_1Q_1^0 + K_2Q_2^0 - K_{1,2}Q_{1,2}^{0,0} + \overline{a_1a_2} \\
&= (c_1 + \widehat{K}_1)Q_1^0 + K_2Q_2^0 - (c_1c_2 + \widehat{K}_{1,2})Q_{1,2}^{0,0} + \overline{a_1a_2} \\
K\widehat{Q} &= \widehat{K}_1\widehat{Q}_1^0 + K_2\widehat{Q}_2^0 - \widehat{K}_{1,2}\widehat{Q}_{1,2}^{0,0} + \overline{a_2}
\end{aligned}$$

Then we have

$$\begin{aligned}
KQ - K\widehat{Q} &= \widehat{K}_1(Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2(Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2}(Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + (c_1Q_1^0 - c_1c_2Q_{1,2}^{0,0} + \overline{a_1a_2} - \overline{a_2}) \\
&= \widehat{K}_1(Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2(Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2}(Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + tb_1S_1^{-1}Q_1^1
\end{aligned}$$

Also

$$\begin{aligned}
\widehat{K}_1(Q_1^0 - \widehat{Q}_1^0) - \widehat{K}_{1,2}(Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) &= \widehat{K}_1b_1Q_1^2 - \widehat{K}_{1,2}b_1Q_{1,2}^{2,0} \\
&= b_1(\widehat{K}_1Q_1^2 - \widehat{K}_{1,2}Q_{1,2}^{0,0} \\
&= b_1(-t\overline{x_1}S_1^{-1}Q_1^2 + t\overline{x_1x_2}Q_{1,2}^{2,0}) \\
&= b_1(t\overline{x_1}S_1^{-1}Q_1^0 - t\overline{x_1x_2}Q_{1,2}^{0,0} - \overline{x_1}Q_1^1)
\end{aligned}$$

Next

$$\begin{aligned}
K_2(Q_2^0 - \widehat{Q}_2^0) &= (c_2 - t\overline{x_1}^2S_2^{-1})\overline{x_2}b_1(Q_2^0 - Q_{1,2}^{0,0}) \\
&= -t\overline{x_1}^2\overline{x_2}b_1S_2^{-1}(Q_2^0 - Q_{1,2}^{0,0}) + c_2b_1\overline{x_1}^2(Q_2^0 - Q_{1,2}^{0,0})
\end{aligned}$$

We can finally combine these expressions into

$$x_1^2(KQ - K\widehat{Q}) = b_1x_1^2(tS_1^{-1}Q_1^1 + t\overline{x_1}S_1^{-1}Q_1^0 - t\overline{x_1x_2}Q_{1,2}^{0,0} - \overline{x_1}Q_1^1 - t\overline{x_1}^2\overline{x_2}S_2^{-1}(Q_2^0 - Q_{1,2}^{0,0}) + c_2\overline{x_1}^2(Q_2^0 - Q_{1,2}^{0,0}))$$

We cancel out the b_1 and we have the equation

$$KQ - K(x_1Q_1^1 + Q_1^0)) = tx_1S_1^{-1}(x_1Q_1^1 + Q_1^0) - (tx_1\overline{x_2}Q_{1,2}^{0,0} + x_1Q_1^1 + t\overline{x_2}S_2^{-1}(Q_2^0 - Q_{1,2}^{0,0})) - c_2(Q_2 - Q_{1,2}^{0,0})$$

Note that

$$K + tx_1S_1^{-1} = 1 - t\overline{x_1}S_1^{-1}$$

Once we expand the KQ term on the LHS we get

$$\begin{aligned}
&c_1 Q_1^0 - t\overline{x_1} S_1^{-1} Q_1^0 + c_2 Q_2^0 - t\overline{x_2} S_1^{-1} Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} + t\overline{x_1 x_2} Q_{1,2}^{0,0} + \overline{a_1 a_2} \\
=&(1 - t\overline{x_1} S_1^{-1})(x_1 Q_1^1 + Q_1^0) - x_1 Q_1^1 + t\overline{x_1 x_2} Q_{1,2}^{0,0} - t\overline{x_2} S_2^{-1} Q_2^0 + c_2 (Q_2^0 - Q_{1,2}^{0,0})
\end{aligned}$$

and after careful book-keeping all the terms cancel out one by one. This completes the long-winded proof of the identity.

Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.